ELVERIQ | Enterprise Warehouse Management Research & Build Reference

Cross-platform competitive feature atlas, 300-capability specification, target architecture, delivery roadmap and validation standards

Product and engineering research dossier

12 November 2026 | Version 1.0

Reference map

This dossier is organized for research review, engineering design and weekly backlog planning. Use the headings in Word navigation or the section titles below.

Executive research conclusion — strategic findings and the gap between transaction recording and physical execution.

1. Scope, method, and evidence rules — how vendor evidence and Elveriq status are classified.

2. Competitive landscape — 17 vendor profiles and a cross-market pattern table.

3. Elveriq repository audit — what the supplied codebase does and does not establish.

4. Proposed product model — architecture, entities, state machines and transaction invariants.

5. Master capability register — 300 numbered feature requirements with testable acceptance criteria across 22 operational areas.

6-8. Process blueprints, interfaces and roadmap — actual warehouse flows, personas and phased outcomes.

9-10. Quality and API contracts — concurrency tests, failure matrix and interface sketches.

11-12. Differentiation and research program — commercially useful focus and how to keep evidence current.

13-15. References, glossary and implementation directive — bibliography, terms and development sequence.

The companion CSV register includes the same 300 EV-NNN identifiers, feature states, priorities, benchmark source IDs and acceptance tests. Start with the document for context; use CSV IDs to create engineering tickets.

Reference status: Research and engineering-planning document, not a live-product certification, licensed vendor comparison, implementation commitment or security assessment. Prepared against the uploaded Elveriq-main.zip repository snapshot and official supplier web materials reviewed for this project.

Mission: Build Elveriq from a transactional warehouse SaaS prototype into a complete, extensible warehouse execution platform, capable of handling real physical workflows as reliably as it handles digital orders.

Executive research conclusion

The defining competitive gap is **execution**. Elveriq today can maintain a tenant, products, warehouses, stock totals, reservations, purchase orders, basic receipts, transfers, counts and returns. Market-leading products provide a second layer that turns customer demand and physical stock into verified work: directing staff to exact bins, scanning the right item or pallet, holding damaged goods, selecting picking tactics, packing cartons, booking carriers, loading trucks and explaining late shipments. A third layer coordinates customers, equipment, workforce, yards, automation and intelligent decisions. Those three layers must share a single consistent inventory truth.

This study benchmarks **17 named commercial systems or suites** and captures **300 candidate capabilities in 22 operational domains**. It is intentionally broad. The list is an applied union of documented capabilities across the vendor set, not an allegation that any single vendor offers all 300 items, that all editions include them, or that every proposed Elveriq implementation already exists in a competitor product. **Each feature entry describes an Elveriq requirement proposed using its cited source as a design reference**. Vendor statements are separated from our proposed acceptance tests. Feature availability, pricing, integration entitlements and service-level commitments must be rechecked before commercial claims.

The repository audit found **12 implemented capabilities, 14 partial capabilities and 274 gaps in this granular benchmark catalog**. These counts reflect the deliberate detail level and are not a maturity score, competitive ranking or percentage of completed work. The functional status refers to the reviewed archive only. A basic warehouse stock balance does not fulfill bin-level, lot-level or pallet-level traceability. A read-only pick list is not warehouse-directed picking. A return receipt is not a full RMA and inspection process. Preliminary repository evidence appears in Section 3.

Decision recommendation. Stabilize financial and inventory correctness first; build location-level inventory + warehouse task orchestration + scanning as one integrated platform; then complete picking, packing and dispatch. Add multi-owner/3PL billing and customer self-service for commercially differentiated growth. Preserve a modular path for enterprise labor, yard, manufacturing and automation. Never substitute a beautiful dashboard for validated physical execution.

Research-at-a-glance

Named platforms or suites studied — **Finding:** 17 (enterprise, midmarket, SMB and 3PL-focused);

Distinct source links — **Finding:** 38 (predominantly vendor-primary documentation and product pages);

Candidate feature specifications — **Finding:** 300 across 22 domains;

Current archive assessment — **Finding:** 12 built; 14 partial; 274 not established in the reviewed repository;

Immediate non-negotiable — **Finding:** PostgreSQL concurrency, migrations, backups, security and pilot controls;

First core product engine — **Finding:** Bin stock + task state machine + scan verification + immutable movement audit;

First full physical outcome — **Finding:** PO or ASN to receiving, putaway, pick, carton pack, stage, load, ship and trace;

Six findings that shape the product

Micro-location inventory is foundational. Odoo, Infor, Microsoft, NetSuite and the enterprise suites distinguish stock in bins, zones, statuses and sometimes handling units. Elveriq's aggregate balance should remain as a reportable roll-up rather than its sole source of physical truth [S01, S10, S21, S22, S24].

Task orchestration is the workflow backbone. Scanners need to confirm an assigned task with a verified source, destination, quantity, owner, lot and operator. The task engine makes directed receiving, putaway, picking, replenishment and counting possible [S02, S22, S26].

Fulfillment is a chain of stateful evidence. Enterprise WMS features include waves, scan-verified picking, packing stations, cartons/pallets, staging, loading and carrier documentation. Elveriq currently handles only early reservation/fulfillment logic [S01, S14, S19, S25].

Commercial 3PL software is a distinct competitive category. Extensiv, ShipHero, Infor and Tecsys document customer portals, multi-owner inventory and event-based service billing. These features create value for a warehouse operator whose customers own the goods [S10, S13, S15, S16, S18].

Advanced vendors now integrate optimization into operations. Blue Yonder's published 2026 features highlight shift forecasting, slotting heatmaps and robotics orchestration; Oracle and Infor describe wave/operations agents. These are valuable advanced layers, but their outputs require correct transaction data and human override [S07, S09, S11, S38].

Cloud trust is a core WMS feature. No operational sophistication compensates for lost inventory, duplicate picks, tenant leakage or a failed restore. The prototype explicitly warns that it is not ready for business-critical inventory. Security and reliability are part of the feature backlog, not a final polish phase.

1. Scope, method, and evidence rules

Scope. A multi-tenant cloud software service for independent warehouses, distributors, e-commerce fulfillment centers and later 3PLs. Elveriq is not itself a freight carrier, retailer, physical warehouse, or custodian of inventory unless future business choices change. We are researching software capabilities relevant to those customer classes; products such as transportation systems or carrier networks may be integrations rather than functionality built natively.

Evidence approach. Vendor-specific observations rely on official feature pages, technical help, learning sites and selected release notes. Sources are catalogued in the bibliography by S-number and URL. Product pages demonstrate that a vendor *advertises* a function; they do not prove availability in every edition or performance in a live deployment. Documentation often changes after a release. Features with no suitable primary source were framed as proposed Elveriq architecture, not assigned as verified competitive claims. Security and standards references come from OWASP, NIST, GS1 US and FDA. The supplied competitor write-up was useful context, but this paper rechecked vendor features rather than treating an earlier unsourced summary as independent proof.

Research limitations. No demo account of each competitor was operated; no private product manual, price quote, implementation contract, production metrics or customer configuration was examined. Some capabilities require partner products, extra modules, devices, implementations, or licenses. Vendor marketing performance figures were excluded from our internal forecasts; real Elveriq performance targets must be measured in pilots. Pricing, packaging and architectural decisions are not guaranteed across vendors. This is a curated 17-vendor study, not literally every WMS vendor in the world. Revisit at least quarterly.

Feature-state rules for Elveriq. "Built" = a relevant component exists in the archive. "Partial" = workflow exists in reduced form, not the accepted advanced capability. "Gap" = no sufficient implementation established in that archive. These are static source-inspection classifications, not comprehensive executed QA results. For each row, an S-reference means a **benchmark or motivating source for the feature area**, not necessarily an exact 1:1 vendor implementation of every design choice in our requirement or acceptance test.

Priority scheme. P0 = pre-pilot integrity, security or retained foundation; P1 = essential end-to-end warehouse execution; P2 = competitive operational and commercial expansion; P3 = high-complexity optimization or

specialized operations; P4 = advanced automation, regulated niches or optional intelligence. Priority is a proposed sequence, not a calendar commitment.

2. Competitive landscape: seventeen platform benchmarks

The selected set includes deep enterprise suites, composable cloud WMS products, midmarket inventory suites and purpose-built e-commerce/3PL systems. We should avoid copying proprietary UI or workflows mechanically. Instead, benchmark outcomes: correct stock; shorter handling time; fewer mispicks; reliable customer billing; and real-time exception resolution.

2.1 SAP Extended Warehouse Management

Position: Enterprise execution ceiling. **Documented footprint:** Deep inbound/outbound, RF, warehouse monitor, lot/serial, labor, waves, HU and yard; integration with SAP ecosystem. **Commercial caution:** Dense configuration and SAP ecosystem dependencies. **What Elveriq should adopt:** Warehouse tasks, handling units, exception command center. **Sources:** [S01], [S02], [S35].

2.2 Manhattan ActiveWarehouse

Position: Unified enterprise operations. **Documented footprint:** Unified inventory, yard, labor, automation; editions designed for operational scale. **Commercial caution:** Edition differences: Essentials vs Enterprise vs Premier. **What Elveriq should adopt:** Unified work + dock + workforce model; modular packaging. **Sources:** [S03], [S04].

2.3 Blue Yonder Warehouse Management

Position: AI-orchestrated warehouse operations. **Documented footprint:** Execution, resource forecasting/orchestration, robotics hub, slotting, returns, analytics and AI. **Commercial caution:** Specific agents, features and packaging vary by version. **What Elveriq should adopt:** Explainable intelligence layered over transactional WMS. **Sources:** [S05], [S06], [S07], [S38].

2.4 Oracle Fusion Cloud Warehouse Management

Position: Enterprise omnichannel / manufacturing. **Documented footprint:** Advanced wave operations, cross-docking, flow-through, VAS, yard and production-linked inventory. **Commercial caution:** Broader Oracle SCM products may be separate. **What Elveriq should adopt:** Retail/wholesale order complexity and manufacturing edge. **Sources:** [S08], [S09].

2.5 Infor WMS

Position: High-configurability, multi-owner WMS. **Documented footprint:** Bin and license plate inventory, voice and RF, QC, 3PL billing, 3D visual warehouse, labor standards. **Commercial caution:** Advanced tools and integrations require fit assessment. **What Elveriq should adopt:** Rules, visual operations and financial customer costing. **Sources:** [S10], [S11].

2.6 Infios Warehouse Management

Position: Modular industrial and automation WMS. **Documented footprint:** Composable labor, voice, WCS, AMR, slotting, yard and warehouse execution. **Commercial caution:** Modules may be licensed separately; focus on deployment needs. **What Elveriq should adopt:** Automation adapters and simulation-ready foundation. **Sources:** [S12].

2.7 Tecsys WMS

Position: Complex distribution and regulated industries. **Documented footprint:** Visual worker guidance, adaptable receiving/picking/replenishment and commercial 3PL services. **Commercial caution:** Regulated customer compliance requires separate validation. **What Elveriq should adopt:** Accessible worker UX and specialist workflows. **Sources:** [S13].

2.8 Deposco Bright Warehouse

Position: Cloud omnichannel fulfillment. **Documented footprint:** Wave and waveless fulfillment, directed replenishment, cartonization, shipping rate shopping and robotics. **Commercial caution:** Marketing service-level figures are not Elveriq forecasts. **What Elveriq should adopt:** Packing to carrier pipeline and workflow orchestration. **Sources:** [S14].

2.9 Extensiv 3PL Warehouse Manager

Position: 3PL commercial reference. **Documented footprint:** 3PL-specific billing, customer portal, scanning, API, EDI, parcel processing, directed putaway. **Commercial caution:** Certain functions belong to additional products or tiers. **What Elveriq should adopt:** Owner-scoped inventory, billable events, portal and self-service. **Sources:** [S15], [S16], [S17].

2.10 ShipHero WMS

Position: High-volume e-commerce / 3PL. **Documented footprint:** Mobile picking, batch automation, barcode validation, 3PL billing, channels, portal and webhooks. **Commercial caution:** Marketing performance and integration counts are not guaranteed. **What Elveriq should adopt:** Fast worker flows and integration-rich onboarding. **Sources:** [S18].

2.11 Logiwa IO

Position: DTC fulfillment optimization. **Documented footprint:** Put-wall, walking-path optimization, carrier shipping, many pick methods and client portals. **Commercial caution:** Optimizations should be benchmarked on real warehouse layouts. **What Elveriq should adopt:** Packing workstations and dense pick strategy library. **Sources:** [S19], [S20].

2.12 Odoo Inventory

Position: Practical SMB functionality baseline. **Documented footprint:** Detailed location rules, FEFO/FIFO, picking strategies, barcodes, reservations, replenishment, return and customer flows. **Commercial caution:** Some features depend on other Odoo apps or configuration. **What Elveriq should adopt:** Minimum usability baseline for serious small warehouses. **Sources:** [S21].

2.13 Microsoft Dynamics 365 SCM WMS

Position: ERP-linked warehouse operations. **Documented footprint:** Configurable mobile tasks, license plates, waves, small parcel, labels, load, MHAX and monitoring. **Commercial caution:** Some mobile/advanced functions may be release- or module-specific. **What Elveriq should adopt:** Work execution state machines and handheld design. **Sources:** [S22], [S23].

2.14 NetSuite WMS

Position: ERP-native midmarket warehouse. **Documented footprint:** Mobile force scan, smart counting, bins, receiving, wave and pick tasks, pack and shipping integration. **Commercial caution:** WMS SuiteApps and dependencies matter. **What Elveriq should adopt:** Inventory-linked wave/task transactions and guided count. **Sources:** [S24], [S25], [S26], [S37].

2.15 Cin7 Core WMS

Position: SMB and inventory-centered operations. **Documented footprint:** Mobile receive, putaway, pick/pack, bins, stocktake, transfers, zone and FIFO/FEFO controls. **Commercial caution:** Advanced functionality differs by plan. **What Elveriq should adopt:** Lean mobile workflows suitable for independent warehouses. **Sources:** [S27].

2.16 Made4net SCExpert

Position: Warehouse + yard + delivery suite. **Documented footprint:** Core execution with labor, yard, last mile, routing and 3PL extensions. **Commercial caution:** Suite components are not all core WMS features. **What Elveriq should adopt:** Modular extension and supply-chain edges. **Sources:** [S28], [S29].

2.17 Fishbowl Advanced Warehouse

Position: SMB distribution and stock control. **Documented footprint:** Barcode/mobile work, multi-warehouse, reorder, 3PL separation and EDI-oriented distribution. **Commercial caution:** Capabilities must be confirmed for purchased version. **What Elveriq should adopt:** Simple warehouse onboarding and stock discipline. **Sources:** [S30].

2.18 Cross-market patterns and implications

Barcode as default interaction — **Repeated competitive signal:** Odoo, NetSuite, Microsoft, ShipHero and 3PL products; **Elveriq implementation implication:** Mobile-first scan flows, mandatory location checks, deliberate offline handling;

Work release instead of flat lists — **Repeated competitive signal:** SAP, NetSuite, Infor and Microsoft; **Elveriq implementation implication:** Task state machine with owners, assignment, SLAs, exceptions and audit;

Location and handling-unit granularity — **Repeated competitive signal:** SAP, Infor, Oracle, Odoo and Microsoft; **Elveriq implementation implication:** Location/HU and lot balance become core physical ledger dimensions;

Mixed fulfillment modes — **Repeated competitive signal:** Infor, Deposco, Logiwa and Odoo; **Elveriq implementation implication:** Single, wave, cluster, batch, zone, case/pallet and put-wall as selectable strategies;

Multichannel integration — **Repeated competitive signal:** ShipHero, Logiwa, Extensiv, Deposco and Oracle; **Elveriq implementation implication:** API gateway, adapters, canonical external IDs, webhooks and outbox;

3PL is a separate operating model — **Repeated competitive signal:** Extensiv, ShipHero, Infor and Tecsys; **Elveriq implementation implication:** Customer-owned stock, client portals, event billing and SLA reporting;

Enterprise extensions — **Repeated competitive signal:** Manhattan, SAP, Blue Yonder, Infios and Made4net; **Elveriq implementation implication:** Modular yard, workforce, MHE/robotics, AI and transportation; avoid early monolith;

Insight must lead to action — **Repeated competitive signal:** Blue Yonder, Manhattan, Oracle and Infor; **Elveriq implementation implication:** KPI card opens corrective workflow, not a static chart;

3. Elveriq repository audit: reality before roadmap

Snapshot basis: The uploaded Elveriq-main.zip archive. It identifies itself as a prototype using a FastAPI web/API layer, SQLAlchemy ORM, SQLite for development and a configured but unproven PostgreSQL production path. It includes vanilla JS/HTML/CSS frontend, automated tests, staff invitations and several warehouse transaction services. Archive labels in earlier documents are not a live deployment check. Do not treat stale sections of ROADMAP.md and PROGRESS.md as more current than feature implementations and RELEASE-16-MILESTONES.md.

Workspace / login / role — **Concrete evidence in archive:** dispatchforge_app/auth.py, web.py, staff_service.py, models.py: Business, User, StaffInvitation; **Interpret carefully:** Staff invitations exist, but email delivery and advanced IAM are absent;

Product and warehouse — **Concrete evidence in archive:** models.py: Warehouse, Product, StockLevel; inventory_service.py; **Interpret carefully:** Inventory is aggregated by warehouse_id and product_id, not by bin, lot, owner or handling unit;

Stock ledger — **Concrete evidence in archive:** models.py: StockMovement; inventory_service.py; **Interpret carefully:** Movement records are present; event sourcing, full posting controls and status breakdowns are not established;

Customer demand — **Concrete evidence in archive:** models.py: CustomerOrder, OrderLine; order_service.py; **Interpret carefully:** Reservation and fulfillment occur but physical pick/pack/load workflow is not modeled;

Purchasing and receiving — **Concrete evidence in archive:** models.py: PurchaseOrder, PurchaseOrderLine; purchasing_service.py; **Interpret carefully:** Partial receiving exists; ASN, QC and directed putaway do not;

Warehouse transfers — **Concrete evidence in archive:** models.py: InventoryTransfer; transfer_service.py; TRANSFERS.md; **Interpret carefully:** Retry-safe transfer foundation; PostgreSQL concurrency tests require execution with live DB;

Counts and returns — **Concrete evidence in archive:** models.py: CycleCount, ReturnReceipt; services and API routes; **Interpret carefully:** Aggregate cycle counts and resellable return intake, not bin counts or full RMA disposition;

Pick list — **Concrete evidence in archive:** picking_service.py; /orders/{id}/pick-list(.csv); **Interpret carefully:** Read-only list, not staff work execution;

Planning and replenishment — **Concrete evidence in archive:** operations_service.py, replenishment_service.py; **Interpret carefully:** Seven-day queue and threshold heuristic, not demand AI or physical pick-face tasks;

Frontend — **Concrete evidence in archive:** templates/dashboard.html, static/app.js, CSS; **Interpret carefully:** Responsive operations UI but no real RF/scanner workflow established;

Operational safety — **Concrete evidence in archive:** SECURITY.md, RELEASE-16-MILESTONES.md, ARCHITECTURE.md; **Interpret carefully:** Prototype explicitly calls for migrations, backup, rate limits, audit, production testing and security review;

Known stock invariant. 0 <= reserved <= on_hand and available = on_hand - reserved. Keep these in the stock accounting model, and expand to a balanced inventory ledger with dimensions for tenant, warehouse, location, owner, product, stock condition, lot, serial and HU as appropriate. Aggregate totals should derive from a single authoritative posting model. Do not create simultaneous unreconciled aggregate and granular ledgers.

Implementation boundary. Functions in inventory.py, warehouse_selection.py and distance.py are legacy educational/simulation material. They do not establish shipping optimization, real warehouse ranking or live network stock in the SaaS application. The repository's own README says no real customer deployment or

production inventory guarantee is established. This paper assumes real user onboarding and live operations are future work.

Source inspection vs testing. We inspected repository files and the declared API/model shapes. We did not verify a hosted environment, paid integrations, scanner devices, real customer logs or all advertised automated tests; do not represent this paper as runtime certification.

4. The proposed Elveriq product model

4.1 Architectural north star

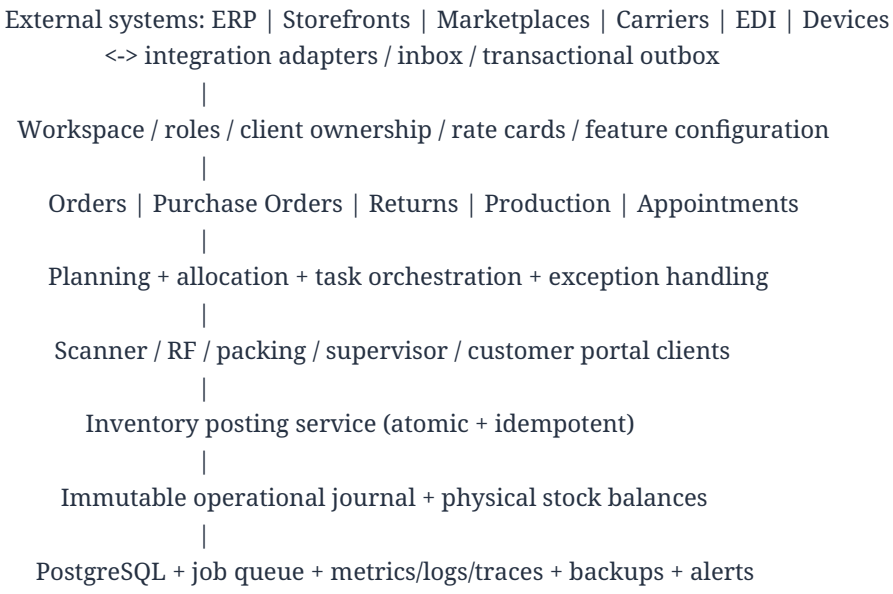
Four layers must remain independently understandable while sharing a single audit trail:

Commercial control plane. Workspaces, contracts, clients, feature editions, users, roles, licensing, invoices and support. One operator workspace may manage multiple customer inventory owners; those concepts are not interchangeable.

Warehouse transaction core. Product catalog, facility graph, ownership, lot/serial, HU, stock ledger, orders, inventory reservations and accounting-safe reconciliation.

Physical execution engine. Work requests, warehouse tasks, workers, scanners, exceptions, receipts, picks, packing stations, shipping loads and completed evidence.

Decision and integration layer. ERP/carrier/e-commerce adapters, events, dashboards, rules, workload optimization, AI, robotics and optional regulated-sector modules.



This is an architecture *proposal*, not a claim that the uploaded repository already has each block. Prefer a modular monolith and explicit service boundaries at first, rather than premature microservices. Split components only when teams, throughput and isolation requirements justify it. The competitor references motivate the feature domains, not the exact technology stack.

4.2 Critical data model additions

StorageLocation — Minimum fields / relationships: tenant, warehouse, zone, parent, code, type, capacity, restriction; **Non-negotiable rule:** code unique within warehouse; hierarchy cannot cycle;

InventoryUnit or balance key — **Minimum fields / relationships:** tenant, warehouse, bin, owner, product, lot, condition, HU, qty; **Non-negotiable rule:** sum to authorized physical stock; no negative available;

Lot / Serial — **Minimum fields / relationships:** product, supplier lot, dates; serial uniqueness domain; **Non-negotiable rule:** preserve receipt-to-shipment genealogy;

HandlingUnit — **Minimum fields / relationships:** barcode/license plate, HU type, parent HU, current bin, owner; **Non-negotiable rule:** no multiple parents or silent merge across owners;

WarehouseTask — **Minimum fields / relationships:** type, work request, source, destination, qty, priority, status, assignee; **Non-negotiable rule:** explicit idempotent state transitions;

TaskEvent / ScanEvent — **Minimum fields / relationships:** actor/device, task, value, timestamp, result, correlation ID; **Non-negotiable rule:** rejected scans audit without stock movement;

ReservationAllocation — **Minimum fields / relationships:** demand line, supply bin, lot/HU, condition, qty, expiration; **Non-negotiable rule:** cannot over-allocate available eligible stock;

ReceivingDelivery / ASN — **Minimum fields / relationships:** PO, supplier, expected line, cartons, shipment, carrier, ETA; **Non-negotiable rule:** discrepancy preserves expected vs actual;

Inspection / Exception — **Minimum fields / relationships:** reason, evidence, hold, owner, approver, lifecycle; **Non-negotiable rule:** hold never automatically becomes sellable;

Package / Shipment / Load — **Minimum fields / relationships:** order lines, HU/carton, label, carrier, staging, load proof; **Non-negotiable rule:** ship only accepted packed quantities;

FulfillmentWave / PickBatch — **Minimum fields / relationships:** selection policy, released tasks, cutoff, status; **Non-negotiable rule:** conserve line quantities through split/merge;

InventoryOwner / ServiceRate / BillableEvent — **Minimum fields / relationships:** 3PL client, contract version, operation, amount/currency; **Non-negotiable rule:** exactly-once event charge and full invoice trace;

IntegrationConnection / InboxEvent / OutboxEvent — **Minimum fields / relationships:** tenant, provider, IDs, payload hash, attempt, status; **Non-negotiable rule:** ingest and publish with idempotency and retry;

DockAppointment / EquipmentResource — **Minimum fields / relationships:** facility, door/asset, capability, schedule, state; **Non-negotiable rule:** no conflicting occupancy or unqualified assignment;

4.3 State machines and transactional truth

Warehouse task: requested -> queued -> released -> assigned -> started -> completed; exceptional branches are blocked, partially completed, cancelled, or reversed with supervisor authorization. “Done” must be caused by scan-confirmed or explicitly supervised evidence, not by navigation or CSV generation.

Inventory: expected (not on hand) -> received/staged (possibly on quality hold) -> bin stock available -> allocated -> picked (in tote/HU) -> packed -> staged -> loaded -> shipped. An inventory state transition should not accidentally double-count a unit in two physical locations. Use one posting API with constrained transitions; reserve a destination and enforce concurrent stock selection where needed.

Outbound order: draft -> validated -> allocated/partially allocated -> released -> picking -> packing -> ready to ship -> staged -> loaded -> shipped; alternate branches: on hold, backordered, cancelled, partially shipped, return pending. Define what actor and event is authorized to cross each boundary. Split shipments and carrier manifest voids require traceable compensating events.

3PL billing: verified operational event -> candidate charge -> rated charge (effective customer contract) -> review -> invoice draft -> approved/exported -> paid/reconciled. Reprocessing a scan must not bill twice. Disputes and credits must not rewrite the source event.

4.4 Concurrency, outbox and idempotency rules

Represent all physical and monetary actions as database transactions. Under concurrent PostgreSQL sessions, two pickers cannot both consume the last available unit in the same bin.

Idempotency keys must scope to tenant and operation, include canonical payload fingerprint, and reject mismatched reuses. A retry returns the original result, never a second receipt or shipment.

Put stock balance updates, task transitions and ledger event creation into a single transaction. Publish integration notifications through a committed outbox, not by calling an external API before commit.

For multi-step work, model explicitly *in transit* or *staged* inventory. Do not reduce warehouse aggregate stock merely because a worker takes goods from one internal aisle to another.

Devices offline for minutes may cache an instruction, but server inventory is authoritative. On reconnection, replay via unique event ID; conflicts require review, not silent overwrite.

Every enterprise KPI must declare source events, formula, timezone, inclusions/exclusions and freshness. An “orders shipped” card reads ship evidence, not orders merely marked fulfilled in a simplistic reservation flow.

5. Master competitive capability register (300 specifications)

Every item below is a proposed Elveriq feature requirement that can be tracked as a backlog issue. The code EV-NNN is stable within this version. “Requirement” describes target behavior; “Acceptance” is a minimum observable test. “References” point to official industry/vendor sources listed in Section 13. The full register is also supplied as CSV for filtering, editing and importing into a ticket system. **Gap** means missing or not established in our inspected archive, not necessarily absent in every live branch.

5.1 SaaS control plane and identity (12 capabilities)

WMS credibility begins with repeatable, secure tenant onboarding, permissions and customer service. **Benchmark references:** [S15], [S16], [S18], [S32], [S33].

EV-001 | Multi-business workspaces

Elveriq: Built | **Priority:** P0 | **References:** [S15], [S18].

Requirement: Keep every company dataset separate, including workers and stock. **Acceptance:** Business A cannot list, change, or infer business B records.

EV-002 | Owner and staff roles

Elveriq: Partial | **Priority:** P0 | **References:** [S15], [S22].

Requirement: Apply centrally defined warehouse permissions per action. **Acceptance:** A picker cannot create users, edit billing, or adjust stock.

EV-003 | Invitation lifecycle

Elveriq: Partial | **Priority:** P0 | **References:** [S18].

Requirement: Invite staff with expiring single-use link and confirmed email delivery. **Acceptance:** Expired and reused links fail and email delivery is audited.

EV-004 | Password recovery and verification

Elveriq: Gap | **Priority:** P0 | **References:** [S33].

Requirement: Provide secure verified reset, recovery, and account verification. **Acceptance:** Tokens expire, cannot be reused, and do not reveal accounts.

EV-005 | Multifactor authentication

Elveriq: Gap | **Priority:** P0 | **References:** [S33].

Requirement: Support authenticator/passkey options for privileged accounts. **Acceptance:** Sensitive actions require appropriate authentication and step-up.

EV-006 | Device and session administration

Elveriq: Gap | **Priority:** P0 | **References:** [S22], [S33].

Requirement: Display and revoke active scanner and web sessions. **Acceptance:** Revoked handheld stops executing tasks promptly.

EV-007 | Role and permission designer

Elveriq: Gap | **Priority:** P1 | **References:** [S22], [S15].

Requirement: Grant scoped actions by warehouse and role, not only owner/staff. **Acceptance:** Policy tests deny unauthorized resource and action combinations.

EV-008 | Warehouse-scoped team assignments

Elveriq: Gap | **Priority:** P1 | **References:** [S22], [S13].

Requirement: A user may work at selected facilities and zones. **Acceptance:** Assignments cannot surface work from an unassigned facility.

EV-009 | Role-based homepage and navigation

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S13].

Requirement: Show relevant actions for receiver, picker, supervisor, owner, 3PL customer. **Acceptance:** Each role sees intended queues, KPI cards, and direct actions.

EV-010 | Customer self-service account

Elveriq: Gap | **Priority:** P2 | **References:** [S16], [S18].

Requirement: Allow 3PL inventory owners to see only their own stock and orders. **Acceptance:** Customer users cannot see another inventory owner's data.

EV-011 | Feature flags and editions

Elveriq: Gap | **Priority:** P2 | **References:** [S03], [S06].

Requirement: Enable modules by edition, customer agreement, and warehouse. **Acceptance:** Disabled features cannot execute by direct API call.

EV-012 | Tenant activation and onboarding checklist

Elveriq: Gap | **Priority:** P1 | **References:** [S15], [S18].

Requirement: Wizard covers facilities, units, employees, inventory import, first receipt. **Acceptance:** New tenant can complete a safe sample workflow end to end.

5.2 Warehouse topology and space (13 capabilities)

A WMS knows not only which warehouse contains stock but the actual location, capacity and operational path. **Benchmark references:** [S01], [S10], [S20], [S21], [S22].

EV-013 | Warehouse list and details

Elveriq: Built | **Priority:** P0 | **References:** [S20], [S21].

Requirement: Retain multihouse facility registry with tenant scope. **Acceptance:** Warehouse records cannot cross tenants.

EV-014 | Warehouse zones and areas

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Define receiving, reserve, pick, pack, returns, quarantine and dispatch areas. **Acceptance:** Each zone has type, identifier, and permissible movements.

EV-015 | Aisles racks shelves bins

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S24].

Requirement: Model hierarchical and unique machine-scannable physical locations. **Acceptance:** One bin belongs to one warehouse and has unique readable code.

EV-016 | Locations and type flags

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Flag reserve vs pick face, floor storage, temperature zone or hazardous restrictions. **Acceptance:** Wrong-zone putaway is blocked by rule.

EV-017 | Put-wall and consolidation cells

Elveriq: Gap | **Priority:** P2 | **References:** [S19].

Requirement: Track staging and put-wall cubbies allocated to each order. **Acceptance:** One cubby never silently mixes orders.

EV-018 | Location capacity dimensions

Elveriq: Gap | **Priority:** P1 | **References:** [S20], [S21].

Requirement: Maintain max units, weight, cube, pallet positions, and compatible SKUs. **Acceptance:** Suggested putaway excludes over-capacity bins.

EV-019 | Location barcode and printable sign

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S25].

Requirement: Print unique shelf/bin identifier that handheld scans must confirm. **Acceptance:** Scan of different bin fails and records incident.

EV-020 | Pick path sequence

Elveriq: Gap | **Priority:** P2 | **References:** [S19], [S22].

Requirement: Define walk order and point-to-point travel graph or sequence. **Acceptance:** Same items yield reproducible accessible route order.

EV-021 | Bin availability and blocks

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S24].

Requirement: Mark bins available, blocked, full, under maintenance or count-locked. **Acceptance:** Work cannot allocate from unavailable locations.

EV-022 | Warehouse occupancy and heatmap

Elveriq: Gap | **Priority:** P3 | **References:** [S07], [S10].

Requirement: Aggregate used volume, travel and replenishment hotspot by zone. **Acceptance:** Utilization calculation reconciles with location stocks.

EV-023 | Multiple storage conditions

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S21].

Requirement: Specify temperature, security or compatibility restrictions by zone. **Acceptance:** Products are routed only to compliant supported locations.

EV-024 | Location audit and genealogy

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Maintain history of moves and adjustments per bin. **Acceptance:** Each on-hand change has actor, source and reason.

EV-025 | Dynamic and fixed pick faces

Elveriq: Gap | **Priority:** P2 | **References:** [S20], [S21].

Requirement: Configure fixed SKU facing or temporarily assign dynamic picking area. **Acceptance:** Allocation respects location mode and replenishment links.

5.3 Product and reference master data (16 capabilities)

Reliable operations require a richer product identity than SKU plus unit price. **Benchmark references:** [S01], [S21], [S22], [S27].

EV-026 | SKU catalog

Elveriq: Built | **Priority:** P0 | **References:** [S21].

Requirement: Preserve tenant-unique products with units and reorder points. **Acceptance:** Duplicate SKU in tenant is rejected.

EV-027 | Supplier directory

Elveriq: Built | **Priority:** P0 | **References:** [S21].

Requirement: Preserve supplier entity and purchasing connections. **Acceptance:** Supplier cannot be linked to other tenant records.

EV-028 | GTIN and alternate barcodes

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S31].

Requirement: Map multiple vendor barcodes and standard GTIN to a SKU. **Acceptance:** A configured barcode resolves deterministically.

EV-029 | Item dimensions and weight

Elveriq: Gap | **Priority:** P1 | **References:** [S14], [S21].

Requirement: Store length, width, height, weight with declared units. **Acceptance:** Cartonization and capacity use converted standard units.

EV-030 | Unit-of-measure conversions

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S10].

Requirement: Allow each, inner, carton, case, pallet with exact conversion ratios. **Acceptance:** Receiving cases preserves base-unit quantity.

EV-031 | Multi-variant product families

Elveriq: Gap | **Priority:** P2 | **References:** [S21].

Requirement: Support size, color and other variation attributes. **Acceptance:** Each sellable variant has separate SKU and inventory.

EV-032 | Lot and batch attributes

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S21].

Requirement: Record batch identity, production dates and supplier lot IDs. **Acceptance:** Every applicable receipt and pick references the correct lot.

EV-033 | Unique serial identification

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S25].

Requirement: Capture serial at receive, move, pack and return. **Acceptance:** One active serial cannot appear in two bins.

EV-034 | Expiration and best-before

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S21].

Requirement: Track sell-by, expiry, shelf-life and minimum remaining life. **Acceptance:** Allocation never violates configured expiry policy.

EV-035 | Catch weight and variable weight

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S10].

Requirement: Track physical units and measured weight when the product requires it. **Acceptance:** Transaction audits both base units and actual weight.

EV-036 | Hazard and handling characteristics

Elveriq: Gap | **Priority:** P3 | **References:** [S22], [S10].

Requirement: Record product segregation, fragility and temperature instructions. **Acceptance:** Incompatible picks and putaway trigger approval or rejection.

EV-037 | Product substitution and cross-reference

Elveriq: Gap | **Priority:** P3 | **References:** [S08], [S21].

Requirement: Define permitted substitutes subject to customer approval. **Acceptance:** Fulfillment uses substitution only within explicit policy.

EV-038 | Kits and BOM product definitions

Elveriq: Gap | **Priority:** P2 | **References:** [S10], [S21].

Requirement: Define multi-component bundles and component quantities. **Acceptance:** Kit availability derives from components without double counting.

EV-039 | Customer-specific product aliases

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S18].

Requirement: Resolve each 3PL customer's item numbers to internal SKU. **Acceptance:** Client mappings never collide across owners.

EV-040 | Vendor pack configuration

Elveriq: Gap | **Priority:** P2 | **References:** [S21], [S27].

Requirement: Track purchase pack sizes, minimum order units and label formats. **Acceptance:** PO and receipt conversions reconcile to base units.

EV-041 | Master data change approval

Elveriq: Gap | **Priority:** P2 | **References:** [S22].

Requirement: Control high-impact edits to product identities and UOM. **Acceptance:** Revision is logged and historical transactions retain identity.

5.4 Inbound planning, ASN and receiving (16 capabilities)

Inbound operations bridge expected supply with real dock receipts, discrepancies and physical units. **Benchmark references:** [S01], [S08], [S10], [S14], [S15].

EV-042 | Purchase orders

Elveriq: Built | **Priority:** P0 | **References:** [S15], [S21].

Requirement: Preserve supplier PO, destination and line quantities. **Acceptance:** Each receipt cannot exceed permitted expected quantity.

EV-043 | Partial receipt against PO

Elveriq: Built | **Priority:** P0 | **References:** [S14], [S15].

Requirement: Preserve incremental receipts with ledger updates. **Acceptance:** Second receipt reflects outstanding balance correctly.

EV-044 | Advance shipping notice ingestion

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S14].

Requirement: Import supplier ASN/856 expected cartons, items and dates. **Acceptance:** Supplier ASN matches PO lines or enters discrepancy review.

EV-045 | Scheduled inbound appointments

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S22].

Requirement: Book carrier, trailer, dock, time and planned handling resources. **Acceptance:** Conflicting dock windows are flagged.

EV-046 | Expected shipment and container

Elveriq: Gap | **Priority:** P2 | **References:** [S08], [S10].

Requirement: Track multiple POs, trailers and containers per arrival. **Acceptance:** Unloaded units tie back to shipment and supplier.

EV-047 | Unplanned inbound receiving

Elveriq: Gap | **Priority:** P2 | **References:** [S10], [S27].

Requirement: Receive authorized unannounced stock with controlled exception. **Acceptance:** Unknown receipt requires supervisor reason and ownership.

EV-048 | Receiving work queue

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S15].

Requirement: Display inbound arrivals and assignment to receiving workers. **Acceptance:** Supervisor can assign, prioritize and track pending receipts.

EV-049 | Scan-verified receiving

Elveriq: Gap | **Priority:** P1 | **References:** [S16], [S25].

Requirement: Scan expected item, lot, quantity and receiving area. **Acceptance:** Unexpected barcode does not enter sellable stock silently.

EV-050 | Blind receiving option

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S25].

Requirement: Hide expected qty for chosen counts to improve independent accuracy. **Acceptance:** Comparison appears after submission, not before.

EV-051 | Over short and damaged receipts

Elveriq: Gap | **Priority:** P1 | **References:** [S15], [S22].

Requirement: Record quantity and damage discrepancy with disposition. **Acceptance:** Cannot over-receive beyond tolerance without approval.

EV-052 | Receiving photograph and attachments

Elveriq: Gap | **Priority:** P2 | **References:** [S12], [S15].

Requirement: Attach evidence to delivery or discrepancy without exposing other tenant. **Acceptance:** File types, sizes and retention are controlled.

EV-053 | Case and pallet label creation

Elveriq: Gap | **Priority:** P1 | **References:** [S14], [S31].

Requirement: Print unique handling-unit label for received packages. **Acceptance:** Each label resolves to package contents and location.

EV-054 | Drop trailer and unloading workflow

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S12].

Requirement: Track trailer check in, seal status, unload start and complete. **Acceptance:** Arrival timestamps and users are auditable.

EV-055 | Vendor performance scorecard

Elveriq: Gap | **Priority:** P3 | **References:** [S13], [S21].

Requirement: Measure PO fill, damage, ASN accuracy and on-time arrivals. **Acceptance:** Report values reconcile with receipts and planned dates.

EV-056 | Direct receipt to cross dock

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S08].

Requirement: Reserve inbound units against outbound demand without reserve storage. **Acceptance:** Outbound demand and receipt are linked with traceable transfer.

EV-057 | Receiving landed-cost reference

Elveriq: Gap | **Priority:** P3 | **References:** [S21], [S23].

Requirement: Attach freight and duties as optional ERP-facing finance data. **Acceptance:** Financial amounts never silently change stock quantity.

5.5 Inspection, quality and quarantine (10 capabilities)

Counted material is not necessarily available for sale. Quality state must be physically and digitally enforced. **Benchmark references:** [S01], [S10], [S21], [S22].

EV-058 | Receiving quality inspection

Elveriq: Gap | **Priority:** P1 | **References:** [S10], [S21].

Requirement: Run inspection checklist before release to available stock. **Acceptance:** Rejected lots cannot be allocated to orders.

EV-059 | Hold and quarantine statuses

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S22].

Requirement: Represent quality, damaged, recall, customs, and customer hold. **Acceptance:** Available-to-pick excludes all nonreleasable stock.

EV-060 | Disposition workflow

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Authorize release, return to vendor, rework or destruction. **Acceptance:** Every status change records approver and reason.

EV-061 | Quality defect codes

Elveriq: Gap | **Priority:** P2 | **References:** [S10], [S15].

Requirement: Maintain searchable standard reasons for defects and shortages. **Acceptance:** Exception trend report groups by code.

EV-062 | Sampling and AQL policy

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S13].

Requirement: Configure product- or supplier-specific sample inspection. **Acceptance:** Failed sample routes lot to quarantine.

EV-063 | Inbound inspection photos

Elveriq: Gap | **Priority:** P2 | **References:** [S12], [S15].

Requirement: Attach evidence with traceable time and actor. **Acceptance:** Documents remain linked after item relocation.

EV-064 | Outbound QC gate

Elveriq: Gap | **Priority:** P2 | **References:** [S25], [S14].

Requirement: Verify order correctness and special handling before handoff. **Acceptance:** Failed check stops carrier manifest completion.

EV-065 | Temperature excursion record

Elveriq: Gap | **Priority:** P4 | **References:** [S13], [S34].

Requirement: Optionally record cold chain excursions and mitigation actions. **Acceptance:** Shipment remains held pending authorized disposition.

EV-066 | Recall search and stock hold

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S21].

Requirement: Identify all on-hand and shipped units sharing affected lot. **Acceptance:** Recall produces an auditable affected-customer report.

EV-067 | Quality dashboards and aging

Elveriq: Gap | **Priority:** P2 | **References:** [S13], [S22].

Requirement: Show held stock value, unresolved exceptions and turnaround time. **Acceptance:** Dashboard reconciles to current hold records.

5.6 Putaway and internal warehouse tasks (12 capabilities)

The task engine turns transaction intent into accountable physical actions. **Benchmark references:** [S01], [S02], [S10], [S21], [S22].

EV-068 | Directed putaway suggestion

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S10].

Requirement: Choose bin using compatibility, space, velocity and strategy. **Acceptance:** Candidate bin is valid at confirmation time.

EV-069 | Putaway scan confirmation

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S25].

Requirement: Require source HU and destination bin scan before completion. **Acceptance:** Wrong bin scan leaves task open and stocks unchanged.

EV-070 | System-directed task object

Elveriq: Gap | **Priority:** P1 | **References:** [S02], [S22].

Requirement: Create task type, source, destination, item, quantity, assignee and status. **Acceptance:** Task has immutable history and replay-safe completion.

EV-071 | Task assignment and release

Elveriq: Gap | **Priority:** P1 | **References:** [S02], [S22].

Requirement: Assign work by role, zone, priority and worker availability. **Acceptance:** One task cannot be simultaneously completed twice.

EV-072 | Task status lifecycle

Elveriq: Gap | **Priority:** P1 | **References:** [S22], [S26].

Requirement: Support queued, released, assigned, started, blocked, done, cancelled. **Acceptance:** Only authorized state transitions are allowed.

EV-073 | Multi-step movement work

Elveriq: Gap | **Priority:** P2 | **References:** [S02], [S22].

Requirement: Allow receive-to-staging-to-storage transitions as linked tasks. **Acceptance:** No missing or duplicated units across chained steps.

EV-074 | Ad-hoc bin to bin move

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S27].

Requirement: Worker may request permitted location move with scan. **Acceptance:** Source quantity decrements as destination increments atomically.

EV-075 | Task split and partial completion

Elveriq: Gap | **Priority:** P2 | **References:** [S22], [S26].

Requirement: Handle shortage, damaged items and residual tasks. **Acceptance:** Completing partial quantity leaves explicit remainder.

EV-076 | Task interleaving

Elveriq: Gap | **Priority:** P3 | **References:** [S02], [S35].

Requirement: Mix putaway and pick work to reduce empty travel. **Acceptance:** Sequence selection is explainable and preserves priorities.

EV-077 | Equipment-aware assignment

Elveriq: Gap | **Priority:** P3 | **References:** [S05], [S12].

Requirement: Require forklift skill, equipment availability or aisle access. **Acceptance:** Unqualified worker cannot accept restricted work.

EV-078 | Task exception and supervisor resolution

Elveriq: Gap | **Priority:** P1 | **References:** [S22], [S19].

Requirement: Block workflow on missing stock, blocked bin or scan mismatch. **Acceptance:** Supervisor can resolve with audit and no phantom inventory.

EV-079 | Warehouse task SLA and escalation

Elveriq: Gap | **Priority:** P2 | **References:** [S05], [S22].

Requirement: Set due-by time and raise overdue notifications. **Acceptance:** Late tasks appear in actionable queue with cause.

5.7 Inventory ledger, reservations and traceability (17 capabilities)

Quantity truth must reconcile at each location, condition, lot, serial and owner. **Benchmark references:** [S01], [S10], [S21], [S22], [S24].

EV-080 | Per-warehouse on hand and reserved

Elveriq: Built | **Priority:** P0 | **References:** [S21].

Requirement: Preserve quantity and reservation invariant at SKU/warehouse level. **Acceptance:** No transaction makes reserved exceed on hand.

EV-081 | Append-only movement history

Elveriq: Built | **Priority:** P0 | **References:** [S21].

Requirement: Preserve user and reason for warehouse aggregate changes. **Acceptance:** Movement history reconciles with aggregate after replay.

EV-082 | Bin-level on hand and reserved

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S10].

Requirement: Add inventory balances keyed by physical bin and stock attributes. **Acceptance:** Sum of bin balances matches warehouse aggregate.

EV-083 | Available, allocated, damaged and QC states

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S22].

Requirement: Separate physical existence from legal availability. **Acceptance:** On-hold units never show as available-to-promise.

EV-084 | Owner and consignment inventory

Elveriq: Gap | **Priority:** P2 | **References:** [S10], [S15].

Requirement: Track economic owner separately from tenant running warehouse. **Acceptance:** Orders allocate only from authorized ownership pools.

EV-085 | Lot genealogy and shipped trace

Elveriq: Gap | **Priority:** P2 | **References:** [S01], [S21].

Requirement: Connect supplier receipt, movements, orders and shipped lots. **Acceptance:** Query reveals all orders shipped from an affected lot.

EV-086 | Serial lifecycle and uniqueness

Elveriq: Gap | **Priority:** P2 | **References:** [S01], [S25].

Requirement: Track serial event path without duplicated active assignment. **Acceptance:** Same serial cannot be booked at two locations.

EV-087 | License plate / handling unit tracking

Elveriq: Gap | **Priority:** P1 | **References:** [S10], [S22].

Requirement: Identify pallets, cartons and totes independent of SKU. **Acceptance:** Moving parent HU moves its contained inventory consistently.

EV-088 | Available to promise ATP

Elveriq: Gap | **Priority:** P2 | **References:** [S08], [S21].

Requirement: Calculate future on hand using inbound and outbound commitments. **Acceptance:** Promise date has traceable supply and demand components.

EV-089 | Reserved stock expiration and release

Elveriq: Gap | **Priority:** P2 | **References:** [S21], [S22].

Requirement: Configure timebound soft reservations and release rules. **Acceptance:** Expired holds free only intended quantity.

EV-090 | Stock ownership transfer

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S15].

Requirement: Record approved change of stock owner while keeping location. **Acceptance:** Both owners receive immutable audit references.

EV-091 | Inter-warehouse stock transfer

Elveriq: Built | **Priority:** P0 | **References:** [S20].

Requirement: Preserve retry-safe completed moves between facilities. **Acceptance:** Retry with same idempotency key doesn't repeat movement.

EV-092 | In-transit inventory lifecycle

Elveriq: Gap | **Priority:** P2 | **References:** [S22], [S29].

Requirement: Separate dispatch and destination receipt for transfers. **Acceptance:** Destination increases only after verified receipt.

EV-093 | Inventory status reclassification

Elveriq: Gap | **Priority:** P1 | **References:** [S22], [S27].

Requirement: Move inventory from inspected to sellable or damaged states. **Acceptance:** Reason and approver are mandatory for restricted statuses.

EV-094 | Stock aging by lot and bin

Elveriq: Gap | **Priority:** P2 | **References:** [S21].

Requirement: Compute receipt age and expiry exposure from movements. **Acceptance:** Report distinguishes FIFO receipt age from last move age.

EV-095 | Negative adjustment approval policy

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Require privileges, reason and thresholds for manual corrections. **Acceptance:** Large write-off waits for approval and is logged.

EV-096 | Cycle count freeze policy

Elveriq: Gap | **Priority:** P2 | **References:** [S24], [S27].

Requirement: Allow count-at-snapshot with safe parallel work handling. **Acceptance:** Detected intervening transaction triggers correct resolution.

5.8 Counting, inventory accuracy and replenishment (13 capabilities)

Control discrepancies and keep pick faces supplied without chasing spreadsheets. **Benchmark references:** [S21], [S22], [S24], [S27].

EV-097 | Aggregate cycle count records

Elveriq: Built | **Priority:** P0 | **References:** [S24].

Requirement: Preserve creation, posting and stale snapshot checks. **Acceptance:** Posting stale count does not overwrite newer stock.

EV-098 | Blind bin cycle count

Elveriq: Gap | **Priority:** P1 | **References:** [S24], [S25].

Requirement: Guide worker to count bins without expected quantity. **Acceptance:** Expected value hidden until submission.

EV-099 | ABC count frequency rules

Elveriq: Gap | **Priority:** P2 | **References:** [S21], [S28].

Requirement: Schedule counts by velocity, value and historical variance. **Acceptance:** Schedule is reproducible and honors cycle cadence.

EV-100 | Triggered exception counts

Elveriq: Gap | **Priority:** P2 | **References:** [S22], [S27].

Requirement: Create count after missing pick or unexpected receive. **Acceptance:** Count closes the triggering exception and records variance.

EV-101 | Location quarantine during count

Elveriq: Gap | **Priority:** P2 | **References:** [S22].

Requirement: Prevent unsafe allocation or allow version-aware adjustment. **Acceptance:** Every overlapping move is reconciled explicitly.

EV-102 | Count variance approvals

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Set quantity/value thresholds and reason taxonomy. **Acceptance:** Unauthorised variance cannot change stock ledger.

EV-103 | Count KPI and root cause

Elveriq: Gap | **Priority:** P2 | **References:** [S13], [S24].

Requirement: Track accuracy, repeat issues and variance by zone/worker. **Acceptance:** Report shows denominator and calculation period.

EV-104 | Threshold-based replenishment

Elveriq: Partial | **Priority:** P1 | **References:** [S21].

Requirement: Preserve reorder-point analysis and open PO consideration. **Acceptance:** Recommendation quantity and existing demand reconcile.

EV-105 | Min/max pick-face replenishment

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Generate replenishment task when pick face below target. **Acceptance:** Replenishment respects reserve-bin and pick-bin quantities.

EV-106 | Demand-triggered replenishment

Elveriq: Gap | **Priority:** P2 | **References:** [S10], [S21].

Requirement: Create task to supply short picking work and due orders. **Acceptance:** Replenishment priority inherits shipment cutoff.

EV-107 | Inter-warehouse reorder suggestions

Elveriq: Gap | **Priority:** P2 | **References:** [S21], [S20].

Requirement: Suggest moves before purchasing when another site has excess. **Acceptance:** Approved transfer respects available stock and transit time.

EV-108 | Supplier lead-time and safety stock

Elveriq: Gap | **Priority:** P2 | **References:** [S21].

Requirement: Model lead time, variability, service level and review cadence. **Acceptance:** Suggested order documents assumptions and units.

EV-109 | Auto draft PO or transfer

Elveriq: Gap | **Priority:** P2 | **References:** [S21].

Requirement: Convert approved rule recommendations to draft source documents. **Acceptance:** No automatic committed purchase without explicit permission.

5.9 Orders, omnichannel and allocation (15 capabilities)

Orders from one or many channels become executable demand with exact promises. **Benchmark references:** [S08], [S14], [S18], [S19], [S21].

EV-110 | Customer order entry

Elveriq: Built | **Priority:** P0 | **References:** [S19], [S21].

Requirement: Preserve multi-line API order, warehouse and due date. **Acceptance:** Order quantities reserve inventory only once.

EV-111 | Scheduled reservation

Elveriq: Built | **Priority:** P0 | **References:** [S21].

Requirement: Preserve safe reserve, fulfill and cancel behavior. **Acceptance:** Cancellation releases reserved not shipped stock.

EV-112 | Order import and validation

Elveriq: Gap | **Priority:** P1 | **References:** [S19], [S18].

Requirement: Import CSV/API orders with mapping, errors and deduplication. **Acceptance:** Duplicate external order is idempotent.

EV-113 | External sales channel connector

Elveriq: Gap | **Priority:** P2 | **References:** [S18], [S20].

Requirement: Pull authorized orders from marketplaces and shops. **Acceptance:** Connector logs every sync and skips already applied events.

EV-114 | Omnichannel order normalization

Elveriq: Gap | **Priority:** P2 | **References:** [S08], [S19].

Requirement: Standardize addresses, units, service, SKU aliases and promises. **Acceptance:** Same order has unambiguous canonical line and state.

EV-115 | Inventory allocation rules engine

Elveriq: Gap | **Priority:** P1 | **References:** [S08], [S21].

Requirement: Allocate by availability, ownership, lot and priority policy. **Acceptance:** No unit is allocated to multiple competing orders.

EV-116 | Partial allocation and backorders

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S26].

Requirement: Ship available quantities and retain remainder with ETA. **Acceptance:** Open backorder reconciles to original order lines.

EV-117 | Future inbound order allocation

Elveriq: Gap | **Priority:** P2 | **References:** [S08], [S21].

Requirement: Optionally reserve expected receipts with cautious promise policy. **Acceptance:** Expected inbound cannot be treated as available stock.

EV-118 | Order priority and cutoffs

Elveriq: Gap | **Priority:** P1 | **References:** [S05], [S19].

Requirement: Set ship-by targets, customer tiers and carrier cutoffs. **Acceptance:** Higher urgency appears ahead in releasable task queue.

EV-119 | Multi-warehouse order sourcing

Elveriq: Gap | **Priority:** P2 | **References:** [S08], [S20].

Requirement: Compare network inventory, shipping distance and split costs. **Acceptance:** Routing explanation shows chosen facility and stock source.

EV-120 | Order hold and cancellation

Elveriq: Partial | **Priority:** P1 | **References:** [S01], [S21].

Requirement: Extend cancel with fraud, payment, customer, QA and address holds. **Acceptance:** Held order cannot be released to picking.

EV-121 | Customer order change management

Elveriq: Gap | **Priority:** P2 | **References:** [S01], [S08].

Requirement: Allow safe address, line and quantity amendments before ship. **Acceptance:** Every change recalculates stock and work dependencies.

EV-122 | Wholesale B2B retail compliance

Elveriq: Gap | **Priority:** P3 | **References:** [S14], [S18].

Requirement: Store retailer routing, labeling, carton and EDI rules. **Acceptance:** Compliance checklist blocks non-conforming shipment.

EV-123 | Order aggregation and split shipments

Elveriq: Gap | **Priority:** P2 | **References:** [S08], [S22].

Requirement: Merge or split fulfillment work without losing source order links. **Acceptance:** All packages trace to correct original demand.

EV-124 | Dropship and direct fulfillment

Elveriq: Gap | **Priority:** P3 | **References:** [S21], [S19].

Requirement: Represent supplier-to-customer dispatch distinct from WMS stock. **Acceptance:** No phantom warehouse movement is recorded.

5.10 Picking and execution strategies (19 capabilities)

This is the operational heartbeat of a serious WMS: verified pick work and adaptable strategies. **Benchmark references:** [S01], [S10], [S19], [S21], [S22], [S26].

EV-125 | Read-only pick list and CSV

Elveriq: Partial | **Priority:** P0 | **References:** [S19], [S26].

Requirement: Preserve printable order item list without claiming execution. **Acceptance:** Printed list agrees with reserved quantities.

EV-126 | Single order directed picking

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S25].

Requirement: Tell picker exact source bin, lot, qty and target container. **Acceptance:** Scan verifies correct item and bin before commit.

EV-127 | Picker work queue

Elveriq: Gap | **Priority:** P1 | **References:** [S22], [S26].

Requirement: Display accepted, assigned, due, and blocked pick tasks. **Acceptance:** Picker cannot complete another worker's restricted tasks.

EV-128 | Scan-to-confirm picks

Elveriq: Gap | **Priority:** P1 | **References:** [S25], [S27].

Requirement: Verify item and source bin or require controlled override. **Acceptance:** Wrong scan cannot ship wrong SKU silently.

EV-129 | Short pick and substitution exception

Elveriq: Gap | **Priority:** P1 | **References:** [S22], [S19].

Requirement: Capture available qty, exception reason and reallocation request. **Acceptance:** No reserved quantity remains locked after final resolution.

EV-130 | Pick reservation at bin level

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Reserve stock from specific lot/bin and release on cancellation. **Acceptance:** Two pickers cannot remove same last unit.

EV-131 | Wave picking

Elveriq: Gap | **Priority:** P2 | **References:** [S01], [S10].

Requirement: Combine due orders into timed release groups by policy. **Acceptance:** Wave generates pick tasks and reports remaining work.

EV-132 | Batch picking

Elveriq: Gap | **Priority:** P2 | **References:** [S19], [S21].

Requirement: Pick shared SKUs for many orders in one trip. **Acceptance:** Sorted units are allocated back to correct orders.

EV-133 | Cluster picking by cart

Elveriq: Gap | **Priority:** P2 | **References:** [S19], [S21].

Requirement: Use one cart with assigned order totes for many orders. **Acceptance:** Every scan goes to correct tote and order.

EV-134 | Zone picking and pick-pass

Elveriq: Gap | **Priority:** P2 | **References:** [S19], [S10].

Requirement: Divide work among zones and merge results at consolidation. **Acceptance:** No order marked ready until all zones complete.

EV-135 | Put-wall / put-to-light

Elveriq: Gap | **Priority:** P3 | **References:** [S19], [S18].

Requirement: Sort a batch to cubbies or light-directed locations. **Acceptance:** A wrong cubby produces an actionable alert.

EV-136 | Pick-to-carton

Elveriq: Gap | **Priority:** P3 | **References:** [S14], [S19].

Requirement: Direct picks straight into designated outbound cartons. **Acceptance:** Carton contents reconcile with packed contents.

EV-137 | Full pallet and case picks

Elveriq: Gap | **Priority:** P2 | **References:** [S19], [S10].

Requirement: Support handling whole case/pallet without individual each scans. **Acceptance:** Movement conserves base-unit quantities.

EV-138 | Pick path optimization

Elveriq: Gap | **Priority:** P3 | **References:** [S19], [S10].

Requirement: Sort pick stops to reduce travel while respecting equipment paths. **Acceptance:** Generated route improves or equals baseline on test layout.

EV-139 | FEFO/FIFO selection enforcement

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S27].

Requirement: Prefer correct stock age/expiry and enforce when required. **Acceptance:** Expired or wrong-lot pick is blocked with reason.

EV-140 | Voice picking interface

Elveriq: Gap | **Priority:** P4 | **References:** [S01], [S10].

Requirement: Enable hands-free guided prompts on supported devices. **Acceptance:** Confirmation logs exact pick with no duplicate posting.

EV-141 | Wearable scanner support

Elveriq: Gap | **Priority:** P4 | **References:** [S22].

Requirement: Support configured glove scanners and scan feedback. **Acceptance:** Unreliable radio retries never duplicate stock moves.

EV-142 | Pick productivity and replenishment coupling

Elveriq: Gap | **Priority:** P2 | **References:** [S05], [S19].

Requirement: Trigger replenishment when pick face becomes insufficient. **Acceptance:** Picking task blocks rather than silently shorting demand.

EV-143 | Re-pick and reversal process

Elveriq: Gap | **Priority:** P2 | **References:** [S22].

Requirement: Reverse errors through permissioned compensating transactions. **Acceptance:** Original pick remains traceable and stock consistent.

5.11 Packing, handling units and VAS (12 capabilities)

Packages and pallets are real warehouse objects, not just printable shipping files. **Benchmark references:** [S01], [S08], [S14], [S19], [S21].

EV-144 | Carton and pallet handling units

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S31].

Requirement: Track package identity, type, hierarchy and current location. **Acceptance:** A pallet's contents remain queryable after a move.

EV-145 | Packing station queue

Elveriq: Gap | **Priority:** P1 | **References:** [S14], [S19].

Requirement: Show picked orders and available packing benches. **Acceptance:** Packing begins only when required pick work is complete.

EV-146 | Scan-to-pack verification

Elveriq: Gap | **Priority:** P1 | **References:** [S25], [S18].

Requirement: Scan picked items into cartons, track short/over pack. **Acceptance:** Carton contents reconcile with order lines.

EV-147 | Multi-carton order support

Elveriq: Gap | **Priority:** P1 | **References:** [S14], [S21].

Requirement: Split an order into many cartons and track each independently. **Acceptance:** Pack quantities sum to order shipped quantities.

EV-148 | Nested handling unit hierarchy

Elveriq: Gap | **Priority:** P2 | **References:** [S01], [S31].

Requirement: Allow items in cartons, cartons on pallets, pallets in staging. **Acceptance:** A unit cannot belong to two parents simultaneously.

EV-149 | Cartonization suggestion

Elveriq: Gap | **Priority:** P3 | **References:** [S14], [S19].

Requirement: Choose carton based on dimensions, weight and fragility rules. **Acceptance:** Recommendation never exceeds physical capacity.

EV-150 | Dunnage and material tracking

Elveriq: Gap | **Priority:** P3 | **References:** [S21], [S22].

Requirement: Record cartons, bags, tape, inserts and costs per package. **Acceptance:** Materials update supplies ledger if configured.

EV-151 | Packing slips and custom documents

Elveriq: Gap | **Priority:** P2 | **References:** [S14], [S19].

Requirement: Generate document templates per account and service. **Acceptance:** Reprinted documents have stable source revision.

EV-152 | GS1 SSCC pallet label

Elveriq: Gap | **Priority:** P2 | **References:** [S31], [S14].

Requirement: Generate logistic-unit IDs and compliant label fields for authorized prefixes. **Acceptance:** Check digit valid and duplicate ID prohibited.

EV-153 | Weight scale confirmation

Elveriq: Gap | **Priority:** P3 | **References:** [S19], [S22].

Requirement: Capture measured package weight and flag variance. **Acceptance:** Weight mismatch pauses release for review.

EV-154 | Kitting and bundle assembly

Elveriq: Gap | **Priority:** P2 | **References:** [S10], [S21].

Requirement: Consume component stock and create traceable assembled item. **Acceptance:** Each kit build balances source and output quantities.

EV-155 | Value-added service work orders

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S10].

Requirement: Track relabeling, bundling, gift wrap, inspection and rework. **Acceptance:** VAS task can generate appropriate billable event.

5.12 Shipping and dispatch (15 capabilities)

Outbound fulfillment is incomplete until actual packed goods are tendered and audited. **Benchmark references:** [S08], [S14], [S15], [S19], [S22].

EV-156 | Shipment record lifecycle

Elveriq: Gap | **Priority:** P1 | **References:** [S08], [S22].

Requirement: Track created, staged, ready, manifested, loaded and shipped states. **Acceptance:** Shipped requires evidence of complete accepted quantities.

EV-157 | Carrier service selection

Elveriq: Gap | **Priority:** P2 | **References:** [S14], [S19].

Requirement: Choose carrier and service by speed, weight and constraints. **Acceptance:** Selection is stored with estimated and final cost.

EV-158 | Multi-carrier rate shopping

Elveriq: Gap | **Priority:** P2 | **References:** [S14], [S16].

Requirement: Compare eligible provider rates using live credentials. **Acceptance:** Chosen rate and unavailable provider errors are auditable.

EV-159 | Parcel labels and tracking IDs

Elveriq: Gap | **Priority:** P1 | **References:** [S14], [S15].

Requirement: Request printable label, store tracking and label revision. **Acceptance:** Retry never buys an unintended second label.

EV-160 | Carrier manifest and closeout

Elveriq: Gap | **Priority:** P2 | **References:** [S14], [S19].

Requirement: Close packages by manifest or end-of-day handoff. **Acceptance:** Manifest totals reconcile with recorded packages.

EV-161 | Address verification

Elveriq: Gap | **Priority:** P2 | **References:** [S19], [S18].

Requirement: Validate destination before rate purchase and print. **Acceptance:** Invalid address forces correction or approval.

EV-162 | Staging lanes by dock and carrier

Elveriq: Gap | **Priority:** P1 | **References:** [S14], [S22].

Requirement: Assign scanned packed units to staging zone. **Acceptance:** Unit cannot be loaded from the wrong lane unnoticed.

EV-163 | Scan-to-load confirmation

Elveriq: Gap | **Priority:** P1 | **References:** [S01], [S22].

Requirement: Verify HUs against truck or shipment at dock. **Acceptance:** Shipment closes only with correct loaded IDs.

EV-164 | Load plan and trailer capacity

Elveriq: Gap | **Priority:** P3 | **References:** [S08], [S12].

Requirement: Assign pallets to loads using weight/cube restrictions. **Acceptance:** Capacity limits show explicit infeasibility.

EV-165 | Bill of lading and retail documents

Elveriq: Gap | **Priority:** P2 | **References:** [S19], [S14].

Requirement: Produce BOL, packing list, ASN/EDI docs when applicable. **Acceptance:** Document totals reconcile with loaded contents.

EV-166 | Proof of delivery integration

Elveriq: Gap | **Priority:** P3 | **References:** [S29].

Requirement: Store recipient signature/event from authorized driver or carrier. **Acceptance:** POD is linked to immutable shipment record.

EV-167 | Shipment exception management

Elveriq: Gap | **Priority:** P1 | **References:** [S08], [S19].

Requirement: Handle short, damaged, carrier error and late cutoff. **Acceptance:** Shipment status reflects unresolved exception.

EV-168 | Partial shipment and backorder

Elveriq: Gap | **Priority:** P1 | **References:** [S08], [S21].

Requirement: Ship available portion while preserving open balance. **Acceptance:** No duplicate reservation or delivery on re-run.

EV-169 | Shipping cost and margin analytics

Elveriq: Gap | **Priority:** P3 | **References:** [S14], [S16].

Requirement: Track cost per parcel/line/customer/channel. **Acceptance:** Reports show which costs are estimates versus billed.

EV-170 | Tracking sync and notifications

Elveriq: Gap | **Priority:** P2 | **References:** [S18], [S19].

Requirement: Publish tracking events and customer communications. **Acceptance:** Duplicate carrier webhook does not duplicate events.

5.13 Returns and reverse logistics (10 capabilities)

Advanced returns make condition, ownership and disposition explicit. **Benchmark references:** [S01], [S08], [S21], [S27].

EV-171 | Restock confirmed customer returns

Elveriq: Partial | **Priority:** P0 | **References:** [S21], [S27].

Requirement: Preserve current return intake limited to confirmed resellable units. **Acceptance:** Duplicate return reference never restocks twice.

EV-172 | RMA authorization

Elveriq: Gap | **Priority:** P2 | **References:** [S21], [S08].

Requirement: Create expected return, reason and return labels before arrival. **Acceptance:** Only authorized qty can be returned without exception.

EV-173 | Return receiving scan

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S27].

Requirement: Scan returned serial/lot/product and inspect condition. **Acceptance:** Return ID ties item to original order if available.

EV-174 | Condition grading

Elveriq: Gap | **Priority:** P2 | **References:** [S21], [S27].

Requirement: Classify new, resellable, damaged, refurbish, scrap and vendor return. **Acceptance:** Only released units increase sellable inventory.

EV-175 | Quarantine by default

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S22].

Requirement: Hold unknown-condition returns until inspected. **Acceptance:** Uninspected return cannot fulfill a customer order.

EV-176 | Rework and refurbishment tasks

Elveriq: Gap | **Priority:** P3 | **References:** [S21], [S10].

Requirement: Generate traceable repair, repack and retest workflow. **Acceptance:** Disposition records before/after state and costs.

EV-177 | Supplier return (RTV)

Elveriq: Gap | **Priority:** P2 | **References:** [S14], [S21].

Requirement: Create return-to-vendor shipment with stock and claim reference. **Acceptance:** Inventory leaves warehouse only on verified issue.

EV-178 | Customer refund link

Elveriq: Gap | **Priority:** P3 | **References:** [S21], [S08].

Requirement: Connect return disposition to approved external financial action. **Acceptance:** Refund is never initiated solely by warehouse scan.

EV-179 | Recall and bulk reverse logistics

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S21].

Requirement: Locate impacted packages and return flow for affected lots. **Acceptance:** Affected stock is held consistently network-wide.

EV-180 | Returns analytics

Elveriq: Gap | **Priority:** P2 | **References:** [S08], [S15].

Requirement: Track reasons, damages, recovery and restock time. **Acceptance:** Metrics reconcile to actual RMA and receipt records.

5.14 3PL customer operations and billing (17 capabilities)

A 3PL WMS manages inventory on behalf of separate customer owners, with accurate charge capture. **Benchmark references:** [S10], [S13], [S15], [S16], [S18].

EV-181 | 3PL customer accounts

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S18].

Requirement: Create inventory-owner entities under each warehouse operator tenant. **Acceptance:** One owner cannot read or allocate another owner's SKU.

EV-182 | Customer-specific warehouses and zones

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S10].

Requirement: Reserve or share locations by agreement and owner rules. **Acceptance:** Shared bins retain ownership at every unit level.

EV-183 | Customer portal inventory

Elveriq: Gap | **Priority:** P2 | **References:** [S16], [S18].

Requirement: Expose scoped balances, lots, locations and order status. **Acceptance:** Customer never sees competing client records.

EV-184 | Customer portal order creation

Elveriq: Gap | **Priority:** P2 | **References:** [S16], [S18].

Requirement: Allow approved clients to submit orders with service rules. **Acceptance:** Portal order uses same validation as WMS order.

EV-185 | Service catalog and rate cards

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S16].

Requirement: Define billable services and rates per customer and effective date. **Acceptance:** Rate version applied at event time is immutable.

EV-186 | Storage billing by pallet/day

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S16].

Requirement: Accrue occupied pallet positions or units daily. **Acceptance:** Accrual reconciles with location and inventory snapshots.

EV-187 | Receiving transaction charges

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S16].

Requirement: Bill by pallet, carton, unit or hour according to contract. **Acceptance:** One physical receipt emits only intended charge once.

EV-188 | Pick, pack and VAS event billing

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S18].

Requirement: Capture verified execution events with rate and quantity. **Acceptance:** Partial/failed tasks do not produce completed-task charge.

EV-189 | Minimums and recurring fees

Elveriq: Gap | **Priority:** P3 | **References:** [S15].

Requirement: Handle minimum monthly spend, account fee and thresholds. **Acceptance:** Recompute invoice deterministically with rate versions.

EV-190 | Carrier markup and pass-through

Elveriq: Gap | **Priority:** P3 | **References:** [S16], [S18].

Requirement: Record freight cost, customer charge and permitted markup. **Acceptance:** Client invoice reveals contractual price behavior.

EV-191 | Billing dispute and adjustments

Elveriq: Gap | **Priority:** P2 | **References:** [S15].

Requirement: Provide audited credit, waiver and dispute workflow. **Acceptance:** Original charge retained after correction.

EV-192 | Invoice generation and tax handoff

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S17].

Requirement: Create reviewed draft invoice and export to accounting. **Acceptance:** Reconciliation shows each line's originating work event.

EV-193 | EDI retail routing compliance

Elveriq: Gap | **Priority:** P3 | **References:** [S15], [S18].

Requirement: Map clients' trading partner label and order requirements. **Acceptance:** EDI failure cannot be silently marked successful.

EV-194 | Client contract and SLA reporting

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S13].

Requirement: Report inbound, fulfillment and accuracy metrics by client. **Acceptance:** Report data never crosses client ownership.

EV-195 | Custom branded labels and slips

Elveriq: Gap | **Priority:** P3 | **References:** [S15], [S18].

Requirement: Apply customer-specific document templates. **Acceptance:** Brand selection keyed to inventory owner not staff choice.

EV-196 | Client visibility notifications

Elveriq: Gap | **Priority:** P2 | **References:** [S16], [S18].

Requirement: Send configured stock, backlog, receipt and shipment updates. **Acceptance:** Each notification has tenant-owner isolation and retry log.

EV-197 | Multi-currency and invoice integration

Elveriq: Gap | **Priority:** P4 | **References:** [S15], [S17].

Requirement: Support explicit currency and tax settings via accounting provider. **Acceptance:** Totals round deterministically and distinguish currencies.

5.15 Labor, resources and shift control (12 capabilities)

Managers require visibility of who should do what, what is blocked and why a shift is falling behind. **Benchmark references:** [S02], [S03], [S05], [S10], [S13].

EV-198 | Worker availability and shift roster

Elveriq: Gap | **Priority:** P2 | **References:** [S05], [S13].

Requirement: Record assigned shifts, zone, skill and equipment availability. **Acceptance:** Task assignment respects availability and permissions.

EV-199 | Labor task clock events

Elveriq: Gap | **Priority:** P2 | **References:** [S10], [S13].

Requirement: Capture start, pause, resume, and finish work time. **Acceptance:** Total effective time excludes configured breaks.

EV-200 | Task skill matrix

Elveriq: Gap | **Priority:** P2 | **References:** [S05], [S12].

Requirement: Model forklift, hazardous goods and restricted zone skills. **Acceptance:** Unsafe task assignment fails policy check.

EV-201 | Resource and equipment registry

Elveriq: Gap | **Priority:** P3 | **References:** [S05], [S12].

Requirement: Track pallet jacks, forklifts, scanner and robot assignments. **Acceptance:** Unavailable asset cannot be assigned to simultaneous tasks.

EV-202 | Worker-directed task prioritization

Elveriq: Gap | **Priority:** P2 | **References:** [S05], [S22].

Requirement: Sort by due time, travel, capability and warehouse policy. **Acceptance:** Reason for chosen next task is visible to supervisor.

EV-203 | Labor engineered standards

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S13].

Requirement: Configure expected time by travel, piece, case and exceptions. **Acceptance:** Actual vs standard report uses correct task classes.

EV-204 | Team and worker productivity

Elveriq: Gap | **Priority:** P2 | **References:** [S10], [S13].

Requirement: Track picks per hour, errors, utilization and travel with context. **Acceptance:** KPI excludes downtime and avoids unsafe quotas by design.

EV-205 | Shift launch dashboard

Elveriq: Gap | **Priority:** P3 | **References:** [S07].

Requirement: Show forecasted workload, headcount gap and priority work. **Acceptance:** Inputs and forecast error visible to supervisors.

EV-206 | Labor forecast intraday

Elveriq: Gap | **Priority:** P4 | **References:** [S07].

Requirement: Predict shift workload based on orders and arrivals. **Acceptance:** Forecast is backtested with confidence and override.

EV-207 | Exception supervisor control room

Elveriq: Gap | **Priority:** P1 | **References:** [S02], [S22].

Requirement: Show blocked pick, receive, load and count work in one queue. **Acceptance:** Every alert links to record and actionable resolution.

EV-208 | Worker training and certification

Elveriq: Gap | **Priority:** P3 | **References:** [S13], [S22].

Requirement: Track approved skills, SOP revisions and acknowledgement. **Acceptance:** Unsafe or untrained workers denied restricted operations.

EV-209 | Operator ergonomics and accessibility

Elveriq: Gap | **Priority:** P2 | **References:** [S22], [S13].

Requirement: Offer large scan targets, audible/visual feedback and accessible flows. **Acceptance:** Critical flow usable with scanner and keyboard.

5.16 Yard, docks and transportation (10 capabilities)

Enterprise users expect coordination between truck arrivals, dock door capacity and warehouse work. **Benchmark references:** [S01], [S08], [S10], [S12], [S22].

EV-210 | Dock door and resource directory

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S10].

Requirement: Define door capabilities, operating hours and receiving/shipping type. **Acceptance:** Incompatible trailer cannot be assigned to door.

EV-211 | Dock appointment scheduler

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S10].

Requirement: Schedule carrier slots and expected load volume. **Acceptance:** Double bookings trigger explicit conflict.

EV-212 | Driver, trailer and vehicle check-in

Elveriq: Gap | **Priority:** P3 | **References:** [S12], [S22].

Requirement: Record actual arrival, appointment and physical trailer ID. **Acceptance:** Check in links to inbound/outbound load.

EV-213 | Yard position tracking

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S12].

Requirement: Track trailer in gate, yard slot, dock and departed. **Acceptance:** Trailer occupancy map matches events.

EV-214 | Trailer seal verification

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S12].

Requirement: Record seal checks and exceptions at unload/load. **Acceptance:** Seal variance blocks release pending sign-off.

EV-215 | Door utilization dashboard

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S22].

Requirement: Show planned vs actual door occupancy and idle time. **Acceptance:** KPIs have correct scheduled/actual denominators.

EV-216 | Dock-to-task coordination

Elveriq: Gap | **Priority:** P3 | **References:** [S08], [S22].

Requirement: Open unload or staging tasks when appropriate arrival is checked in. **Acceptance:** Wrong shipment cannot auto-trigger a receiving task.

EV-217 | Cross-dock lane planning

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S08].

Requirement: Reserve staging slots for immediate outbound demand. **Acceptance:** Inbound and outbound linkage visible in real time.

EV-218 | Transportation load tender status

Elveriq: Gap | **Priority:** P4 | **References:** [S08], [S29].

Requirement: Capture booked, dispatched and tendered freight outcomes. **Acceptance:** Failed tender does not show shipped.

EV-219 | Yard alert and detention exposure

Elveriq: Gap | **Priority:** P4 | **References:** [S12], [S29].

Requirement: Alert on trailer dwell time and missed dock slots. **Acceptance:** Dwell reports distinguish carrier vs warehouse delay.

5.17 Automation, equipment and digital workers (11 capabilities)

Automation is optional yet needs safe interoperability, observability and failure handling from day one of design. **Benchmark references:** [S01], [S05], [S06], [S12], [S14], [S22].

EV-220 | Robotics integration gateway

Elveriq: Gap | **Priority:** P4 | **References:** [S01], [S05].

Requirement: Use vendor-neutral command and event adapter for AMRs. **Acceptance:** Adapter failure leaves reversible WMS task state.

EV-221 | Conveyor and sorter integration

Elveriq: Gap | **Priority:** P4 | **References:** [S12], [S22].

Requirement: Exchange material flow events with WCS/MHE safely. **Acceptance:** Messages are ordered or idempotently reconciled.

EV-222 | Automated storage retrieval interface

Elveriq: Gap | **Priority:** P4 | **References:** [S01], [S12].

Requirement: Send retrieval requests and handle equipment exceptions. **Acceptance:** Unavailable device does not lose reserved inventory.

EV-223 | Pick-to-light interface

Elveriq: Gap | **Priority:** P4 | **References:** [S14], [S18].

Requirement: Show and confirm pick quantities on supported hardware. **Acceptance:** Misconfirmed quantity is traceable to pick task.

EV-224 | Voice-directed workflow adapter

Elveriq: Gap | **Priority:** P4 | **References:** [S01], [S10].

Requirement: Convert WMS tasks to voice instruction and confirmation. **Acceptance:** Every accepted voice command performs validated task step.

EV-225 | Scale, dimensioner and print services

Elveriq: Gap | **Priority:** P3 | **References:** [S14], [S22].

Requirement: Capture validated measurements and printer job results. **Acceptance:** Device data has timestamp, unit and source identity.

EV-226 | RFID portal event processing

Elveriq: Gap | **Priority:** P4 | **References:** [S02], [S19].

Requirement: Deduplicate reads and corroborate physical movement events. **Acceptance:** Repeated tag reads do not create phantom shipments.

EV-227 | Offline-first handheld queue

Elveriq: Gap | **Priority:** P3 | **References:** [S21], [S22].

Requirement: Allow safe local step capture during short network interruptions. **Acceptance:** Reconnection replays without duplicate stock changes.

EV-228 | Automation simulator and digital twin

Elveriq: Gap | **Priority:** P4 | **References:** [S12], [S10].

Requirement: Test bottlenecks and equipment events using synthetic orders. **Acceptance:** Simulation never modifies production stock.

EV-229 | Manual override and failsafe

Elveriq: Gap | **Priority:** P3 | **References:** [S12], [S22].

Requirement: Offer permissioned emergency stop and alternate manual execution. **Acceptance:** System recovers from robot outage without double execution.

EV-230 | Equipment failure event monitoring

Elveriq: Gap | **Priority:** P4 | **References:** [S05], [S12].

Requirement: Classify faults and reroute tasks to manual queues. **Acceptance:** Active stock and task ledgers remain consistent.

5.18 Dashboards, analytics and operational intelligence (16 capabilities)

The command center should display live actionable facts, not disconnected decorative tiles. **Benchmark references:** [S02], [S05], [S07], [S13], [S16], [S22].

EV-231 | Basic overview KPIs and seven-day planning

Elveriq: Partial | **Priority:** P1 | **References:** [S13], [S22].

Requirement: Preserve current aggregate counts and simple queue. **Acceptance:** KPIs reconcile to current orders and stock ledger.

EV-232 | Open task operational command center

Elveriq: Gap | **Priority:** P1 | **References:** [S02], [S22].

Requirement: Show work by stage, SLA, warehouse and owner with drill-down. **Acceptance:** Every card opens a correctly filtered action list.

EV-233 | Inbound receiving performance

Elveriq: Gap | **Priority:** P2 | **References:** [S13], [S16].

Requirement: Monitor ASN compliance, dwell, discrepancies and receipts per hour. **Acceptance:** Numbers reconcile with posted receiving work.

EV-234 | Outbound throughput and backlog

Elveriq: Gap | **Priority:** P1 | **References:** [S14], [S19].

Requirement: Track orders due, late, lines picked, packed and shipped. **Acceptance:** Status filters match immutable shipment events.

EV-235 | Inventory accuracy by count cycle

Elveriq: Gap | **Priority:** P2 | **References:** [S13], [S24].

Requirement: Show variance count, value, SKU and location scope. **Acceptance:** Accurate denominators and exclusions documented.

EV-236 | Labor productivity dashboards

Elveriq: Gap | **Priority:** P2 | **References:** [S05], [S10].

Requirement: Show tasks/hour, pick errors and utilization with context. **Acceptance:** No fabricated utilization when time clock unavailable.

EV-237 | Warehouse space utilization

Elveriq: Gap | **Priority:** P3 | **References:** [S07], [S10].

Requirement: Measure occupied positions and cube by zone and aisle. **Acceptance:** Aggregate reconciles with bin occupancy.

EV-238 | Cost-to-serve by customer

Elveriq: Gap | **Priority:** P3 | **References:** [S13], [S15].

Requirement: Calculate work minutes, packaging and freight expense by client. **Acceptance:** Costs link to billable events and actual resources.

EV-239 | Client 3PL billing dashboards

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S16].

Requirement: Show unbilled work, invoice amount and disputes. **Acceptance:** Charges can be traced to verified operations.

EV-240 | SLA breach prediction

Elveriq: Gap | **Priority:** P3 | **References:** [S05], [S07].

Requirement: Flag likely missed cutoffs using remaining work and capacity. **Acceptance:** Every prediction includes risk driver and historical precision.

EV-241 | Alert routing and escalation

Elveriq: Gap | **Priority:** P2 | **References:** [S02], [S22].

Requirement: Route operational exceptions by shift and urgency. **Acceptance:** Alert acknowledgement and resolution are auditable.

EV-242 | Custom report builder

Elveriq: Gap | **Priority:** P3 | **References:** [S13], [S16].

Requirement: Enable safe curated dimensions and exports by role. **Acceptance:** Cross-tenant joins are impossible in generated queries.

EV-243 | Scheduled exports

Elveriq: Gap | **Priority:** P2 | **References:** [S16], [S21].

Requirement: Send approved report via scoped link or destination. **Acceptance:** Exports honor role, audit and retention policy.

EV-244 | Network-wide multiwarehouse view

Elveriq: Gap | **Priority:** P3 | **References:** [S03], [S20].

Requirement: Roll up facilities while preserving drilldown and owner filter. **Acceptance:** Totals equal sum of selected sites.

EV-245 | Near-real-time dashboard updates

Elveriq: Gap | **Priority:** P2 | **References:** [S05], [S22].

Requirement: Push or poll new events with freshness indicator. **Acceptance:** User can distinguish stale sync from zero activity.

EV-246 | KPI definition catalog

Elveriq: Gap | **Priority:** P1 | **References:** [S13], [S35].

Requirement: Document formulas, units, timezone, cutoffs and source tables. **Acceptance:** Each KPI has automated reconciliation test.

5.19 AI, optimization and decision support (11 capabilities)

AI belongs on top of reliable event data and must be measurable, explainable and reversible. **Benchmark references:** [S03], [S05], [S07], [S09], [S11], [S38].

EV-247 | Slotting ABC recommendations

Elveriq: Gap | **Priority:** P3 | **References:** [S05], [S21].

Requirement: Recommend fast movers near shipping and maintain constraints. **Acceptance:** Recommendations disclose input velocity and constraints.

EV-248 | Slotting heatmap

Elveriq: Gap | **Priority:** P3 | **References:** [S07].

Requirement: Show travel time and slot deviation over facility map. **Acceptance:** Heatmap is supported by actual paths and timestamps.

EV-249 | Picker route optimizer

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S19].

Requirement: Suggest sequence minimizing weighted travel under zone rules. **Acceptance:** Route is benchmarked against simple serpentine baseline.

EV-250 | Workload and shift forecast

Elveriq: Gap | **Priority:** P3 | **References:** [S07].

Requirement: Forecast expected line/task volumes from recent actual arrivals. **Acceptance:** Backtest includes mean absolute error and coverage.

EV-251 | Dynamic labor rebalancing

Elveriq: Gap | **Priority:** P4 | **References:** [S05], [S07].

Requirement: Suggest staff reassignment during bottlenecks. **Acceptance:** Supervisor can accept, reject and audit suggestion.

EV-252 | Warehouse operations assistant

Elveriq: Gap | **Priority:** P4 | **References:** [S05], [S09].

Requirement: Answer natural-language questions with linked source records. **Acceptance:** Assistant never executes write operation without confirmation.

EV-253 | Explainable wave advisor

Elveriq: Gap | **Priority:** P4 | **References:** [S09], [S11].

Requirement: Analyze wave failure and recommend alternative grouping. **Acceptance:** Advice identifies measured cause and forecast impact.

EV-254 | Replenishment forecast suggestions

Elveriq: Gap | **Priority:** P3 | **References:** [S21], [S07].

Requirement: Estimate shortfalls and proposed reorder quantities. **Acceptance:** Model performance compared with threshold baseline.

EV-255 | Anomaly detection in stock movements

Elveriq: Gap | **Priority:** P3 | **References:** [S05], [S07].

Requirement: Flag unusual adjustments, inventory shrink or repeated short picks. **Acceptance:** Alerts show evidence and false positive tracking.

EV-256 | Capacity and simulation workbench

Elveriq: Gap | **Priority:** P4 | **References:** [S12], [S07].

Requirement: Simulate shift workload, resources, picking policies and peak days. **Acceptance:** Sandbox results cannot alter real order state.

EV-257 | Guardrails and AI evaluation registry

Elveriq: Gap | **Priority:** P2 | **References:** [S05], [S09].

Requirement: Track prompt/tool policy, version, evaluation and human approval. **Acceptance:** No unapproved model output changes inventory.

5.20 Integrations, APIs and interoperability (16 capabilities)

Modern WMS must fit into customers existing ERP, e-commerce and carrier infrastructure. **Benchmark references:** [S01], [S08], [S15], [S18], [S19], [S22].

EV-258 | Tenant-scoped REST API

Elveriq: Partial | **Priority:** P0 | **References:** [S15], [S17].

Requirement: Extend existing APIs with documented scopes, versions and pagination. **Acceptance:** Auth and tenant boundary tests pass on all resources.

EV-259 | API keys OAuth and scopes

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S18].

Requirement: Issue rotating credentials with fine-grained least privilege. **Acceptance:** Revoked token cannot access warehouse APIs.

EV-260 | Inbound webhooks

Elveriq: Gap | **Priority:** P2 | **References:** [S18], [S19].

Requirement: Accept signed events from connected systems with replay protection. **Acceptance:** Duplicate webhook is idempotently processed.

EV-261 | Outbound webhooks and outbox

Elveriq: Gap | **Priority:** P1 | **References:** [S15], [S22].

Requirement: Publish inventory and order events only after commit. **Acceptance:** Retry and reconnect cannot lose or duplicate state.

EV-262 | Integration job monitoring

Elveriq: Gap | **Priority:** P2 | **References:** [S17], [S22].

Requirement: Track sync offset, errors, rate limits and retry DLQ. **Acceptance:** Failed messages are visible and safely replayable.

EV-263 | Shopify integration

Elveriq: Gap | **Priority:** P2 | **References:** [S18], [S20].

Requirement: Ingest authorized orders and publish inventory on chosen source-of-truth rules. **Acceptance:** Repeated shop updates cannot create negative stock.

EV-264 | Amazon marketplace integration

Elveriq: Gap | **Priority:** P3 | **References:** [S18], [S19].

Requirement: Use authorized marketplace APIs for orders and shipping data. **Acceptance:** External IDs and cutoff rules are recorded.

EV-265 | TikTok and other channels

Elveriq: Gap | **Priority:** P3 | **References:** [S18].

Requirement: Offer modular channel connector with normalized events. **Acceptance:** Connector can be disabled without stopping core WMS.

EV-266 | QuickBooks accounting connector

Elveriq: Gap | **Priority:** P2 | **References:** [S15], [S17].

Requirement: Export approved invoices or purchasing accounting entries. **Acceptance:** Financial and inventory totals reconcile by reference.

EV-267 | NetSuite/ERP connector

Elveriq: Gap | **Priority:** P3 | **References:** [S18], [S22].

Requirement: Synchronize item master, POs, orders and receipts with ownership rules. **Acceptance:** No infinite echo loop between systems.

EV-268 | EDI 850 856 940 945 846 mapping

Elveriq: Gap | **Priority:** P3 | **References:** [S15], [S18].

Requirement: Support trading-partner profiles and controlled translation. **Acceptance:** Test documents validate partner requirements and acknowledgements.

EV-269 | Carrier and postal APIs

Elveriq: Gap | **Priority:** P2 | **References:** [S14], [S19].

Requirement: Integrate labels, rates, tracking, void and manifests. **Acceptance:** Provider failure is retry-safe and observable.

EV-270 | GS1 and barcode parsing

Elveriq: Gap | **Priority:** P1 | **References:** [S21], [S31].

Requirement: Parse configured GS1 application identifiers with validation. **Acceptance:** GTIN, lot, date and SSCC are not confused.

EV-271 | Print and document service

Elveriq: Gap | **Priority:** P2 | **References:** [S22], [S14].

Requirement: Template ZPL/PDF labels by printer and package type. **Acceptance:** Label generation is idempotent and document audit stored.

EV-272 | Bulk data import and export

Elveriq: Partial | **Priority:** P1 | **References:** [S16], [S21].

Requirement: Extend existing CSV tools to map, preview, reconcile and undo. **Acceptance:** Import rejects malformed rows with error workbook/report.

EV-273 | Partner integration sandbox

Elveriq: Gap | **Priority:** P3 | **References:** [S15], [S22].

Requirement: Provide test tokens, sample payloads, status page and docs. **Acceptance:** Test callbacks never touch production tenant inventory.

5.21 Security, reliability and operational governance (17 capabilities)

A high-feature-count prototype is not a trustworthy WMS until financial and physical transactions survive real conditions. **Benchmark references:** [S22], [S32], [S33].

EV-274 | Tenant authorization regression suite

Elveriq: Partial | **Priority:** P0 | **References:** [S32].

Requirement: Extend tested tenant scoping to every new endpoint and parent-child reference. **Acceptance:** IDOR and privilege escalation tests fail closed.

EV-275 | PostgreSQL transaction validation

Elveriq: Partial | **Priority:** P0 | **References:** [S22].

Requirement: Run concurrent reserve, receipt, transfer and picking tests on PostgreSQL. **Acceptance:** Race tests preserve invariants under retries and crashes.

EV-276 | Versioned schema migrations

Elveriq: Gap | **Priority:** P0 | **References:** [S22].

Requirement: Use Alembic forward and backward versioned migrations. **Acceptance:** Deploy and rollback tested from previous release.

EV-277 | Backup and point-in-time recovery

Elveriq: Gap | **Priority:** P0 | **References:** [S22].

Requirement: Schedule encrypted backups and perform restore drills. **Acceptance:** Defined RPO/RTO demonstrated in timed test.

EV-278 | Structured audit events

Elveriq: Partial | **Priority:** P0 | **References:** [S21], [S32].

Requirement: Record stock actions, roles, actor, reason and correlation ID. **Acceptance:** Auditor can reconstruct who changed inventory and when.

EV-279 | Rate limiting and abuse prevention

Elveriq: Gap | **Priority:** P0 | **References:** [S32].

Requirement: Limit login, exports, automation calls and bulk writes. **Acceptance:** Abusive request pattern is blocked and audited.

EV-280 | Encryption and secret governance

Elveriq: Partial | **Priority:** P0 | **References:** [S33].

Requirement: Protect secrets, sensitive fields and TLS in all environments. **Acceptance:** Keys rotate and no secret appears in logs.

EV-281 | Session revocation and MFA

Elveriq: Gap | **Priority:** P0 | **References:** [S33].

Requirement: Provide administrator invalidation and step-up for high-risk actions. **Acceptance:** Revoked tokens remain unusable across devices.

EV-282 | Monitoring and alerting

Elveriq: Gap | **Priority:** P0 | **References:** [S22].

Requirement: Record uptime, traces, errors, queue lag and inventory anomalies. **Acceptance:** Operational pages identify failing job and affected tenant.

EV-283 | Disaster recovery game day

Elveriq: Gap | **Priority:** P1 | **References:** [S22].

Requirement: Practice restore, worker failover and data reconciliation. **Acceptance:** Recovery produces documented actual downtime and data loss.

EV-284 | Idempotency on stock mutations

Elveriq: Partial | **Priority:** P0 | **References:** [S22].

Requirement: Generalize transfer retry rules to receiving, picks, webhooks and shipments. **Acceptance:** Same idempotency key never executes physical transaction twice.

EV-285 | Event outbox and dead-letter jobs

Elveriq: Gap | **Priority:** P1 | **References:** [S22].

Requirement: Atomically persist events and deliver with safe retries. **Acceptance:** Fault injection does not lose committed event.

EV-286 | Audit retention and privacy policy

Elveriq: Gap | **Priority:** P0 | **References:** [S32], [S33].

Requirement: Set data retention, export, deletion and tenant offboarding rules. **Acceptance:** Policies have tested access and redaction behavior.

EV-287 | Least-privilege and security review

Elveriq: Gap | **Priority:** P0 | **References:** [S32], [S33].

Requirement: Review API permissions and isolate roles, DB and queues. **Acceptance:** Independent findings resolved before production pilot.

EV-288 | Load and large-order performance

Elveriq: Gap | **Priority:** P1 | **References:** [S22].

Requirement: Benchmark realistic SKU counts, multi-line orders, concurrent scanners. **Acceptance:** Service targets met with documented workload.

EV-289 | Feature rollback and safe rollout

Elveriq: Gap | **Priority:** P1 | **References:** [S03], [S22].

Requirement: Use migration-safe rollout flags and recovery playbooks. **Acceptance:** A disabled feature neither loses data nor breaks core workflow.

EV-290 | Operational support and change history

Elveriq: Gap | **Priority:** P1 | **References:** [S16], [S22].

Requirement: Provide ticket-linked audit, incident status and release notes. **Acceptance:** Support can investigate without impersonating clients.

5.22 Vertical specializations and optional extensions (10 capabilities)

Do not imply sector-specific regulatory compliance merely because a generic tracking feature exists. **Benchmark references:** [S01], [S08], [S13], [S22], [S34].

EV-291 | Food lot traceability and recall exports

Elveriq: Gap | **Priority:** P3 | **References:** [S34], [S01].

Requirement: For applicable customers capture traceability events and lot genealogy. **Acceptance:** Export matches configured critical event and element rules.

EV-292 | FEFO and shelf-life customer policy

Elveriq: Gap | **Priority:** P2 | **References:** [S21], [S27].

Requirement: Apply minimum shelf life per customer and order destination. **Acceptance:** Picking refuses lot with insufficient remaining days.

EV-293 | Pharma serial verification adapter

Elveriq: Gap | **Priority:** P4 | **References:** [S13], [S01].

Requirement: Design serial and event boundaries for regulated supply chains. **Acceptance:** Do not claim DSCSA compliance before legal/partner validation.

EV-294 | Cold chain monitoring integration

Elveriq: Gap | **Priority:** P4 | **References:** [S13], [S22].

Requirement: Capture device evidence and hold affected inventory on excursions. **Acceptance:** Exception cannot be auto-cleared without rule and approver.

EV-295 | Manufacturing raw material staging

Elveriq: Gap | **Priority:** P3 | **References:** [S01], [S08].

Requirement: Direct materials to production lines with lot traceability. **Acceptance:** Consumption and return reconcile with warehouse stock.

EV-296 | Kitting to stock and to order

Elveriq: Gap | **Priority:** P2 | **References:** [S01], [S10].

Requirement: Support component pick, assembly and serial association. **Acceptance:** Finished kits trace to exact source components.

EV-297 | Wholesale customer pallet labels

Elveriq: Gap | **Priority:** P3 | **References:** [S14], [S31].

Requirement: Support retailer-compliant logistics labels and shipping documentation. **Acceptance:** Required label data validates before print.

EV-298 | Heavy/oversized item putaway

Elveriq: Gap | **Priority:** P3 | **References:** [S13], [S10].

Requirement: Restrict equipment routes, floor bins and loading techniques. **Acceptance:** Oversize task requires suitable resource and location.

EV-299 | Apparel style color size matrix

Elveriq: Gap | **Priority:** P3 | **References:** [S10], [S21].

Requirement: Support variant attribute hierarchy and bulk order interfaces. **Acceptance:** Search and exports preserve variant identity.

EV-300 | Hazmat operational flags

Elveriq: Gap | **Priority:** P4 | **References:** [S15], [S22].

Requirement: Represent customer rules, segregated storage and shipping constraints. **Acceptance:** No safety/regulatory certification claim without expert review.

6. End-to-end reference process blueprints

The sequences below are operational design hypotheses inspired by the WMS sources. They are not direct descriptions of any one vendor's proprietary internal process. Every step must be traceable to a tenant and warehouse, actor/system, physical unit and auditable time.

6.1 Supplier-to-shelf: purchase order to available bin

Trigger: approved PO or advance shipping notice expects 120 cartons at Facility A. **Sequence:** appointment check-in (if enabled) -> receive work created -> dock staging lane selected -> cartons/HUs scanned -> received quantities, lots and serials verified -> shortages/damage exceptions captured -> quality holds separated -> putaway rules propose valid bin -> worker scans carton/HU and destination bin -> physical location movement posted -> inventory becomes available according to quality disposition -> dashboards and customer/ERP integration events update. **Critical failure branches:** duplicate scan, overreceipt tolerance, invalid SKU alias, full bin, wrong lot, expired item, Wi-Fi loss. **Acceptance:** no unit appears as available before the required receive/inspection/putaway conditions; quantity in all bins plus staging reconciles with receipt ledger and transfers. [S01, S10, S14, S15, S21, S25]

6.2 Order-to-truck: order import to carrier handoff

Trigger: retail or e-commerce order arrives with ship-by cutoff. **Sequence:** ingest/dedupe -> validate item/customer/channel -> allocate eligible inventory by owner, lot/expiry and location -> create work request -> choose single/batch/wave/cluster/zone strategy -> assign tasks -> scan source bin/item/qty -> resolve short pick or replenishment -> direct picked HUs to pack station -> scan to pack cartons -> compute weight/labels/carrier service -> create shipment -> stage at correct dock -> scan cartons/pallets onto load -> confirm departure -> transmit tracking/ASN/EDI if configured. **Failure branches:** order hold after wave release, shortage, wrong carton, printer failure, carrier API timeout, missing loaded pallet. **Acceptance:** order quantity = shipped + explicitly unshipped/backordered/cancelled quantities; all shipped units have a chain of custody through the task and HU ledger. [S08, S14, S18, S19, S22, S26]

6.3 Cycle count without corrupting active stock

Trigger: ABC count plan, exception short pick or supervisor request. **Sequence:** choose bin/products -> select snapshot/freeze policy -> issue blind count task -> worker scans correct location, lot and quantity -> detect overlapping moves -> review threshold variance -> supervisor posts authorized adjustment -> create immutable ledger entry -> update count accuracy KPI. **Failure:** simultaneous pick or receive changes on-hand after snapshot. **Acceptance:** no stale count overwrites current stock, and no negative available stock is introduced. Preserve Elveriq's existing aggregate stale-snapshot protection but adapt it to each location balance. [S21, S24, S25, S27]

6.4 3PL client order plus charge capture

Trigger: Client B (inventory owner) submits order using portal/API. **Sequence:** validate client permissions -> resolve client SKU aliases -> allocate only B-owned inventory -> pick, pack, label, ship -> emit verified billable events for unit/case handling, packaging, storage days and VAS -> rate through effective contract -> review invoice -> export to accounting -> show shipment, stock and charges to B portal only. **Critical risk:** physically commingled identical SKUs for different owners. **Acceptance:** cross-client read and allocation tests deny access; identical actions replayed cannot generate duplicate billable lines. [S10, S15, S16, S18]

6.5 Recall and quarantine

Trigger: vendor or customer flags impacted lot. **Sequence:** identify affected lots/serials -> find all warehouse bins, HUs, orders and shipped customers -> block current inventory -> stop unfinished picks and pack/load release -> export relevant trace results -> initiate approved return/destruction/rework -> preserve event-level evidence. **Acceptance:** confirmed affected stock cannot be allocated while held; released and shipped lineage remain reconstructable. Sector-specific trace requirements are customer- and jurisdiction-dependent; legal advice is needed before compliance claims. [S01, S21, S34]

6.6 3PL seasonal peak shift

Trigger: demand doubles ahead of a cutoff. **Sequence:** receive order forecast and due lanes -> surface work by cutoff, location, skills and equipment -> supervisor changes priorities -> select appropriate wave/batch release -> task router assigns work -> dashboard shows blocker/replenishment and labor deviations -> supervisor resolves exceptions -> after shift, report actual travel, throughput, on-time performance and rate of errors. **Acceptance:** replan preserves all original order commitments and explains changed assignments. AI advice is optional and must not auto-post inventory. [S03, S05, S07, S10, S11]

7. User experience and application map

Avoid a vast cluttered navigation menu. Expose a role-based workspace. Desktop supervisor applications and scanner worker applications are different interfaces sharing the same tasks and posting services.

Business owner / administrator — **Primary workspace:** Business console; **Essential actions and evidence:** Onboarding, facilities, staff, subscription/contract, integration, KPI, audit;

Warehouse operations manager — **Primary workspace:** Operations command center; **Essential actions and evidence:** Backlog, due shipments, blocked work, productivity, inventory accuracy, task assignment;

Receiving worker — **Primary workspace:** Inbound handheld / tablet; **Essential actions and evidence:** ASN/PO lookup, scan, inspect, print HU, directed putaway, photo exception;

Forklift / putaway operator — **Primary workspace:** Material-movement handheld; **Essential actions and evidence:** Assigned HU moves, destination scans, replenishment, blocked task resolution;

Picker — **Primary workspace:** Pick handheld; **Essential actions and evidence:** Next task, source bin, item/lot, exact qty, tote, short pick, route;

Packer — **Primary workspace:** Packing workstation; **Essential actions and evidence:** Order/tote scan, cartonize, scale, QC, labels, release to staging;

Shipping lead — **Primary workspace:** Dispatch console; **Essential actions and evidence:** Staging lanes, load scans, manifests, BOL, exceptions, departure;

Inventory controller — **Primary workspace:** Stock accuracy console; **Essential actions and evidence:** Counts, holds, lot genealogy, variance approval, replenishment and adjustment;

3PL client — **Primary workspace:** Customer portal; **Essential actions and evidence:** Authorized owned inventory, order upload, tracking, SLAs, invoices and disputes;

Integration operator — **Primary workspace:** Integration console; **Essential actions and evidence:** Partner tokens, sync status, failed jobs, webhooks, resync and replay;

Suggested top-level navigation: My Work; Operations; Inbound; Stock & Locations; Outbound; Returns & Quality; Workforce; Yard; Clients & Billing; Insights; Integrations; Administration. Hide optional areas by subscription,

facility configuration and permission. A tap on an overdue alert must lead to the exact source order/task rather than a generic report.

Scanning design doctrine: full-screen confirmation, large readable location code, audio/vibration feedback, ergonomic step order, no hidden data entry, explicit conflict resolution, resilient network reconnect, disabled dangerous actions until server confirmation. Work states must not be inferred from the user closing a screen. Accessibility is a measurable requirement, not a decoration.

8. Delivery strategy and dependency-ordered roadmap

A large feature catalogue is only useful if engineers can build it without fragmenting stock truth. The roadmap below groups outcomes by dependency, not by number of UI screens. The phase timings are intentionally unspecified until engineering capacity and a pilot partner are known.

Gate 0: production safety and honest operational baseline (P0)

Deliver: versioned DB migrations, PostgreSQL load/concurrency tests, backups and restore drills, isolated tenant policies, authentication recovery and MFA plan, rate limiting, secrets handling, structured logs, observability, CI regression and a safe release/rollback procedure. Preserve the existing business, product, PO, order, count, return and transfer flows. **Exit evidence:** realistic simultaneous purchase/receipt/reservation/transfer tests produce the expected stock ledger after retry and failure. A clean encrypted backup is restored and totals reconcile. Do not onboard business-critical stock before these gates are complete.

Gate 1: warehouse spatial model and task core (P1)

Build order: location hierarchy -> physical balances keyed by bin/lot/owner/status -> stock posting API -> task schema/state machine -> worker/zone queue -> scan verifier -> directed receiving/putaway -> location transfer -> exception resolution -> reconciliation/report. **Exit demo:** receive a pallet, label it, inspect it, scan it to a compatible bin, transfer it, count it, and trace every operation from audit to UI; two workers cannot consume the same units simultaneously.

Gate 2: complete outbound physical execution (P1 to P2)

Build order: allocations at bin level -> picking tasks -> short-pick flows -> pick-face replenishment -> batch/wave/cluster -> carton/HU and packing bench -> scan-to-pack -> carrier integration -> staging/load -> tracking and external notification. **Exit demo:** multi-line customer order with lot rules is picked by staff, split across two cartons, labeled, staged, scan-loaded and shipped; quantity and tracking reconcile. Both negative and exception cases pass.

Gate 3: competitive SMB and 3PL commercial surface (P2)

Build order: external order import and API authentication -> marketplace sync with clear stock source of truth -> client inventory owners -> customer portal -> chargeable event engine -> rate cards and contract versions -> invoice drafts and accounting export -> SLA reporting. **Exit demo:** two 3PL customers hold identical SKUs but cannot see, pick or bill each other's goods; client charges originate from verified events and survive replays.

Gate 4: advanced execution and optimization (P3)

Build order: higher-order waves, put-wall, dynamic slotting, supervisor workload plans, labor standards, ASN/dock schedule, cross-dock, VAS, retail compliance, advanced traceability, space analytics and simulation. **Exit**

demo: a complex peak-shift operating scenario can be replanned without losing inventory correctness and with auditable reasons for changed tasks.

Gate 5: enterprise extensions and validated intelligence (P4)

Explore only with suitable customers: robotics adapters, conveyors/MHE, yard/transport integration, voice/RFID, manufacturing, pharmaceutical or cold-chain verticals, AI slotting and labor predictions. **Exit demo:** safe hardware integration with timeouts and operator fallback; measured improvement over baseline; sector-specific external review where compliance is claimed.

Backlog ordering logic

For each candidate feature score **customer pain (0-5)**, **frequency (0-5)**, **risk reduction (0-5)**, **commercial differentiation (0-5)**, **implementation complexity (1-5)** and **dependency readiness (0-5)**. Use a transparent initial score: 2*pain + frequency + 2*risk_reduction + differentiation + dependency_readiness - 2*complexity. This is an internal ranking *proposal*, not a market-measured coefficient. Always override scoring when safety, financial integrity or regulatory obligations demand earlier work. Attach evidence from an interview, pilot or customer ticket.

9. Quality engineering and test catalog

9.1 Invariants that must never be broken

Every InventoryUnit can belong to exactly one current physical storage position or explicitly modeled transit stage; pending delivery forecasts are not physical inventory.

The aggregate by warehouse/product/owner/status/lot reconciles with all current location balances and handling units.

available >= 0 unless a deliberate business adjustment model explicitly accepts controlled negative stock; the initial WMS should not.

All state-changing endpoints enforce tenant boundary, warehouse permission, actor authorization and idempotency key where relevant.

Reservations plus verified picks and other commitments cannot allocate more eligible units than exist.

Every physical movement has an immutable journal entry and, if scanned, source scan evidence and destination confirmation.

Shipment content and customer order fulfilled quantities reconcile with packed and loaded HUs, with explicit exceptions for partially shipped orders.

A 3PL operational event creates at most one intended charge for a particular service rate; a correction creates a credit or reversal record rather than altering history.

Integration retries, server restarts, device reconnects and timeout races cannot duplicate an inventory posting.

9.2 Test matrix by failure mode

Two pickers claim one final unit — **Test method:** Parallel PostgreSQL transactions; **Required result:** One succeeds; the other sees explicit shortage/blocked task;

Wrong bin scan — **Test method:** Handheld integration test; **Required result:** Task unconfirmed; no silent stock move;

Duplicate ASN or carrier webhook — **Test method:** Idempotency and replay test; **Required result:** One effective transaction, auditable duplicate event;

Serial appears in two cartons — **Test method:** Unique serial check; **Required result:** Second pack rejected unless documented correction;

Expired lot picked to restricted customer — **Test method:** Rule engine test; **Required result:** Ineligible lot not allocated or scanned;

Uninspected returned item reserved — **Test method:** Stock-status authorization; **Required result:** Hold excludes it from ATP and pick;

Worker from another tenant guesses order ID — **Test method:** Object-level authorization; **Required result:** No data or mutation; audit relevant denial;

Count overlaps receipt — **Test method:** Snapshot conflict simulation; **Required result:** Variance resnapshot/review; no stale overwrite;

Carrier request times out after label purchased — **Test method:** Network chaos test; **Required result:** Retry finds original label, no duplicate charge;

Wi-Fi drops during pick confirmation — **Test method:** Offline replay test; **Required result:** Acknowledgment reconciles once at reconnect;

Transfer ships but destination not received — **Test method:** Two-stage transit test; **Required result:** Stock appears in transit, not destination usable;

3PL client portal leaks another owner — **Test method:** Contract test across tenant-owner hierarchy; **Required result:** Strict owner isolation;

Backup restore after large shift — **Test method:** Disaster recovery drill; **Required result:** RPO/RTO verified; reconciled stock ledger;

Wave cancelled mid-pick — **Test method:** State transition test; **Required result:** Picked goods remain accountable, unpicked reservations released correctly;

9.3 Test pyramid and pilot strategy

Unit: allocation selectors, lot/FEFO, UOM math, charge rating, invariant checkers and task state transitions. **Service/API:** tenant and owner boundaries, CSRF/session, role checks, transactional posting, revisions, webhooks and bulk imports. **Database:** PostgreSQL serializability/read-committed behavior under actual concurrent requests; isolation and retry. **Browser / handheld:** scanner simulation across receive, count, pick, pack and ship; keyboard accessibility and device rotation. **Integration:** contract tests against sandbox carrier/ERP APIs, signature validation and dead-letter replay. **Load:** synthetic stress resembling a real design partner's SKU/order throughput rather than an invented target. **Pilot:** a consenting warehouse with sanitized initial data, staff training, daily reconciliations, observed time-on-task and documented rollbacks.

Launch acceptance metrics: choose actual targets after pilot baseline. Recommended measured metrics are inventory location accuracy, pick accuracy, stock reconciliation variance, order cycle time, dock-to-stock duration, on-time ship rate, return inspection time, 3PL invoice disputes, support tickets per thousand transactions, API error rate, delayed outbox events and restore effectiveness. Do not promise vendor marketing percentages as Elveriq service guarantees.

10. Engineering work packages and core API contracts

The following interface sketches are planning examples. Endpoint naming, versioning and event schemas must be decided in API design review. Do not use this section as an assertion that these endpoints already exist.

10.1 Location and stock API

POST /api/v1/locations, GET /api/v1/locations/{id}/stock, POST /api/v1/inventory/movements, GET /api/v1/trace/lots/{lot_id}. A stock movement request must identify source location (if any), destination, item, quantity/UOM, owner, stock condition, lot/serial/HU, actor, reason, work reference, tenant and idempotency token. Permission checks and inventory capacity validation are server-side. Avoid accepting arbitrary client-supplied balance totals.

10.2 Work orchestration API

POST /api/v1/work/requests, GET /api/v1/work/tasks?assignee=me, POST /api/v1/work/tasks/{id}/claim, POST /api/v1/work/tasks/{id}/confirm-step, POST /api/v1/work/tasks/{id}/exception. Response includes expected scan type, expected bin/product/HU, quantity, allowed overrides and task revision. The confirmation handler validates source, destination, lot and concurrency before ledger posting. “Claim” is atomic and may include a lease or revision strategy to prevent conflicting supervisors.

10.3 Receiving, outbound and shipping API

POST /api/v1/asns, POST /api/v1/receipts, POST /api/v1/waves, POST /api/v1/packages, POST /api/v1/shipments/{id}/labels, POST /api/v1/loads/{id}/scan. Each API must have explicit lifecycle constraints. Receipt, pack and load confirmation are not interchangeable. Label creation may be a separate external side effect with a transaction/outbox and cancellation/void model.

10.4 Billable event API

POST /api/v1/billing/events should ordinarily be internal or protected from arbitrary client charges. It references the validated physical event, service category, customer owner, rate card version, quantity, unit and correlation ID. Events settle into rated charges and invoices with immutable audit. Reference counts and decimals must not use floating-point arithmetic for currency; store integer minor units or exact decimals with currency controls.

10.5 Canonical event envelope

```
{
 "event_id": "evt_...unique...",
 "event_type": "warehouse.task.completed",
 "schema_version": 1,
 "occurred_at": "2026-11-12T17:20:00Z",
 "tenant_id": "...",
 "warehouse_id": "...",
 "actor_id": "...",
 "aggregate_id": "...",
 "aggregate_version": 5,
 "correlation_id": "...",
 "data": {"task_id": "...", "quantity": 4, "unit": "each"}
}
```

This is an illustrative schema with synthetic identifiers and timestamp, not live warehouse data. Define event ownership, dedupe, redaction, retention, compatibility, schema evolution and replay semantics before using it in production.

11. Competitive differentiation that is achievable

To be taken seriously, Elveriq cannot merely claim “everything”. It should deliver an end-to-end verified workflow for its chosen initial buyers before broadening. Large enterprise products are feature-rich but often require significant design and configuration; modern small-warehouse products put speed and mobility first. Elveriq’s opportunity is a **modular, intuitive execution system with enterprise-grade audit and practical pricing**. Pricing itself is outside this paper because current customer willingness to pay has not been validated.

First target buyer: independent distributor or warehouse with one to three sites and recurring physical receiving/picking problems. **Second target buyer:** small-to-mid-sized 3PL with multiple client inventory owners and billing friction. **Later enterprise buyers:** multi-site operators requiring custom ERP rules, sophisticated labor and automation interfaces. Do not promise unsupported regulated use cases.

Suggested differentiators: fast setup with importer preview; task-first UX rather than menu-first ERP; traceable, truthful KPI cards; configurable workflow templates rather than thousands of default controls; excellent mobile scan feedback; strong exception handling; visible client-level service profitability; and high quality support and training material. Each differentiator needs pilot evidence, not slogans.

Avoid product debt: Do not build five independent inventory counters for picking, returns, HU tracking, ERP sync and dashboard metrics. Avoid premature AI chat that suggests or executes unsafe stock corrections. Do not make a “quick ship” button bypass packing audit. Do not represent a CSV import as a bidirectional integration. Do not copy vendors’ proprietary graphics or code; implement interoperable, independently designed patterns.

12. Operating research program and research questions

Keep this document alive. On every vendor release, add or amend source links, evidence date, feature status, edition requirement, proposed acceptance and competitor screenshot/test notes (with permission). On every Elveriq pull request, link to one or more EV-NNN records and update classification only after verified tests. Every pilot incident becomes an exception-handling acceptance test.

Questions to ask actual warehouse supervisors: Which scans are required today? How are damaged/short receipts handled? Which WMS does the customer use? How much stock is held on the floor versus bins? Are pallet labels and SSCC used? How often does customer inventory mix in one facility? How are carrier manifests produced? What causes daily rework? Which KPIs do managers trust? What features are unused even in their current enterprise WMS? What compliance obligations come from contractual customers rather than law? Which hours of operation and Wi-Fi failure modes must be supported?

Vendor verification template: vendor, product, edition, feature, source, dated capture, observed user flow, limits, external integrations, configuration required, license dependency, hardware requirements, security/control implications, Elveriq design translation and acceptance tests. Never make a “feature parity” claim without a transparent scope and test procedure.

13. Source bibliography and vendor documentation

Links are direct official sources whenever possible. The label [Sxx] in a feature requirement is an evidence trail and a design prompt. It does **not** imply that the cited vendor ships the exact custom Elveriq acceptance test wording or includes that capability in every edition. Sources should be recrawled and saved in an internal research repository when later editions are reviewed.

- **[S01] SAP — SAP EWM features.** https://www.sap.com/canada/products/scm/extended-warehouse-management/features.html. *Evidence note:* Official feature overview; scope varies by deployment and license.
- **[S02] SAP — SAP Learning: EWM processes.** https://learning.sap.com/courses/exploring-business-processes-in-sap-ewm-for-sap-s-4hana-cloud-private-edition/introducing-sap-s4-hana-extended-warehouse-management. *Evidence note:* Warehouse monitor, RF, VAS, labor, slotting, cross-docking.
- **[S03] Manhattan Associates — ActiveWarehouse platform and editions.** https://www.manh.com/solutions/supply-chain-management-software/activewarehouse. *Evidence note:* Inventory, labor, automation, yard; feature differences across editions.
- **[S04] Manhattan Associates — Supply Chain Execution 26.2.** https://www.manh.com/releases/2026/waypoints-supply-chain-execution-26.2. *Evidence note:* Unified warehouse, yard, labor, slotting and AI release capabilities.
- **[S05] Blue Yonder — Warehouse Management Solution.** https://blueyonder.com/solutions/warehouse-management. *Evidence note:* Warehouse execution, resource orchestration, slotting, robotics, analytics.
- **[S06] Blue Yonder — WMS FAQ.** https://blueyonder.com/resources/blue-yonder-ai-powered-wms-unified-faq. *Evidence note:* Features may vary by package, edition and release.
- **[S07] Blue Yonder — 2026.2 cognitive execution release.** https://blueyonder.com/release-announcements/2026-2-release/cognitive-solutions-for-execution. *Evidence note:* Shift forecasts, labor insight, slotting heatmaps.
- **[S08] Oracle — Fusion Cloud Warehouse Management.** https://www.oracle.com/scm/logistics/warehouse-management/. *Evidence note:* Waving, cross docking, VAS, manufacturing and yard operations.
- **[S09] Oracle — WMS AI agent innovations.** https://blogs.oracle.com/scm/log-ai-innovations-26a. *Evidence note:* Wave research and task prioritization agents.
- **[S10] Infor — Infor WMS capability overview.** https://www.infor.com/solutions/scm/warehouse-management-system. *Evidence note:* Multi-owner, bin, wave, voice, 3PL billing, labor, 3D.
- **[S11] Infor — October 2026 supply chain AI agents.** https://www.infor.com/blog/supply-chain-ai-agents-october-2026-release. *Evidence note:* Wave and slotting agents; release-specific.
- **[S12] Infios — Infios WMS capabilities.** https://www.infios.com/en/supply-chain-solutions/warehouse-management-systems. *Evidence note:* Modular labor, yard, voice, WCS, robotics and execution.
- **[S13] Tecsys — Tecsys WMS.** https://www.tecsys.com/solutions/warehouse-management. *Evidence note:* Visual Logistics, warehouse tasks, replenishment and billing.
- **[S14] Deposco — Bright Warehouse WMS.** https://deposco.com/solutions/supply-chain-execution/warehouse-management/. *Evidence note:* Waved and waveless, ASN, scanning, shipping, automation.
- **[S15] Extensiv — 3PL Warehouse Manager product.** https://www.extensiv.com/3pl/products/3pl-warehouse-manager. *Evidence note:* 3PL billing, REST API, ASN, picking, scanning, lot/serial.
- **[S16] Extensiv — 3PL solutions.** https://www.extensiv.com/solutions/3pls. *Evidence note:* Customer portal, SmartScan, rate shopping, analytics, dock.
- **[S17] Extensiv — 3PL knowledge base.** https://help.extensiv.com/en_US/3pl-warehouse-manager. *Evidence note:* Operational submodules, shipping, EDI, 3PL billing and scanning.
- **[S18] ShipHero — Warehouse management software.** https://www.shiphero.com/warehouse-management-software. *Evidence note:* Client billing, portals, barcode verification and channel integrations.

- **[S19] Logiwa — Outbound operations.** https://www.logiwa.com/solutions/fulfillment-of-orders-outbound-operations. *Evidence note:* Pick/pack/ship, put-wall, many pick methods, integrations.
- **[S20] Logiwa — Digital warehouse software.** https://www.logiwa.com/solutions/digital-warehouse-management-software. *Evidence note:* Location rules, capacities, mobile scanners, e-commerce sync.
- **[S21] Odoo — Inventory detailed feature catalog.** https://www.odoo.com/app/inventory-features. *Evidence note:* Location, putaway, picking, lot/serial, FEFO, portals, replenishment.
- **[S22] Microsoft — Dynamics 365 Warehouse Management overview.** https://learn.microsoft.com/en-us/dynamics365/supply-chain/warehousing/warehouse-management-overview. *Evidence note:* Work rules, mobile scanning, load handling, label printing, MHAX.
- **[S23] Microsoft — Dynamics 365 SCM documentation.** https://learn.microsoft.com/dynamics365/supply-chain. *Evidence note:* Module overview including QA, transportation, mobile and planning.
- **[S24] NetSuite — NetSuite WMS features.** https://www.netsuite.com/portal/products/erp/warehouse-fulfillment/wms.shtml. *Evidence note:* Mobile WMS, smart counting, bins, waves.
- **[S25] NetSuite — NetSuite mobile WMS.** https://www.netsuite.com/portal/products/erp/warehouse-fulfillment/mobile-warehouse-management.shtml. *Evidence note:* Force scan verification, RF receiving, counting and picking.
- **[S26] Oracle NetSuite — Releasing orders to warehouse.** https://docs.oracle.com/en/cloud/saas/netsuite/ns-online-help/section_1548078373.html. *Evidence note:* Wave release and pick-task tracking.
- **[S27] Cin7 — Cin7 Core WMS documentation.** https://help.core.cin7.com/hc/en-us/articles/9034461577487-Introduction-to-Warehouse-Management-System-WMS. *Evidence note:* Mobile picking, FIFO/FEFO validation, packing, counting, scanning.
- **[S28] Made4net — WMS core and extended capabilities.** https://made4net.com/knowledge-center/warehouse-order-fulfillment-software-guide/. *Evidence note:* Practical core vs. extended scope; cross-dock, WMS, labor, yard.
- **[S29] Made4net — Supply chain execution suite.** https://made4net.com/. *Evidence note:* WMS, 3PL, yard, labor, routing, proof of delivery.
- **[S30] Fishbowl — Advanced Warehouse.** https://www.fishbowlinventory.com/warehouse-management/. *Evidence note:* Multi-warehouse, mobile, 3PL, EDI, stock controls.
- **[S31] GS1 US — What is SSCC and logistics label.** https://www.help.gs1us.org/barcodes-available-in-gs1-us-data-hub. *Evidence note:* SSCC is logistic-unit identity; not GTIN.
- **[S32] OWASP — API Security Top 10 2023.** https://owasp.org/projects/api-security-project. *Evidence note:* Object level and function level authorization risks.
- **[S33] NIST — Digital Identity Guidelines 800-63B-4.** https://csrc.nist.gov/pubs/sp/800/63/b/4/final. *Evidence note:* Authentication and authenticator management guidance.
- **[S34] FDA — Food Traceability Rule and effective timetable.** https://www.fda.gov/food/food-safety-modernization-act-fsma/fsma-final-rule-requirements-additional-traceability-records-certain-foods. *Evidence note:* Traceability for covered foods; enforcement timing jurisdiction dependent.
- **[S35] SAP — What is a WMS?.** https://www.sap.com/products/scm/extended-warehouse-management/what-is-a-wms.html. *Evidence note:* Labor, yard, monitoring, KPIs and task interleaving.
- **[S36] NetSuite — Warehouse processing help.** https://docs.oracle.com/en/cloud/saas/netsuite/ns-online-help/section_N2317586.html. *Evidence note:* Barcode, bin management, assembly, counts.

- **[S37] NetSuite — WMS overview.** https://docs.oracle.com/en/cloud/saas/netsuite/ns-online-help/chapter_156382517509.html. *Evidence note:* Availability/requirements of WMS SuiteApps; packaging caveat.
- **[S38] Blue Yonder — 2026.3 execution release.** https://blueyonder.com/release-announcements/2026-3-release/cognitive-solutions-for-execution. *Evidence note:* Multi-vendor automation integration and explainable slotting.

14. Glossary of critical warehouse terms

WMS — **Meaning in this paper:** Warehouse Management System: inventory location plus directed execution, counting, shipping and visibility;

WES — **Meaning in this paper:** Warehouse Execution System: real-time operational orchestration that can sit within/alongside a WMS;

WCS — **Meaning in this paper:** Warehouse Control System: automation device or material-handling control interface;

ERP — **Meaning in this paper:** Enterprise Resource Planning; product, purchasing, finance and supply chain source systems;

3PL — **Meaning in this paper:** Third-Party Logistics; a warehouse operator serves inventory-owning customers;

PO — **Meaning in this paper:** Purchase Order; expected supplier order;

ASN — **Meaning in this paper:** Advance Shipping Notice; advance receipt details for an arriving shipment;

HU / LPN — **Meaning in this paper:** Handling Unit / License Plate Number; uniquely identified carton, pallet or tote;

SSCC — **Meaning in this paper:** Serial Shipping Container Code; GS1 identifier for a logistic unit, distinct from product GTIN;

GTIN — **Meaning in this paper:** Global Trade Item Number; identity of a trade item rather than a shipment unit;

RF — **Meaning in this paper:** Radio Frequency; handheld workflow using scanner and mobile device;

FIFO / FEFO — **Meaning in this paper:** First In First Out / First Expiring First Out stock-removal logic;

Wave — **Meaning in this paper:** Released work group of one or many outbound orders;

Put-wall — **Meaning in this paper:** Consolidation/sortation cells used after bulk or batch picking;

Cross-docking — **Meaning in this paper:** Moving inbound goods to outbound demand with minimal long-term storage;

Slotting — **Meaning in this paper:** Planned allocation of SKU locations to improve travel and space use;

ATP — **Meaning in this paper:** Available to Promise; calculation of eligible inventory against commitments and future supply;

VAS — **Meaning in this paper:** Value-Added Services such as kitting, labeling, relabeling and repacking;

MHE — **Meaning in this paper:** Material Handling Equipment (conveyors, sorters, AS/RS, etc.);

AMR — **Meaning in this paper:** Autonomous Mobile Robot;

EDI — **Meaning in this paper:** Electronic Data Interchange; structured trading-partner documents such as orders and shipping notices;

RMA / RTV — **Meaning in this paper:** Return Merchandise Authorization / Return to Vendor;

RPO / RTO — **Meaning in this paper:** Recovery Point Objective / Recovery Time Objective for disaster recovery;

SOP — **Meaning in this paper:** Standard Operating Procedure;

Idempotency — **Meaning in this paper:** Repeated same business command has one intended effective result;

Transactional outbox — **Meaning in this paper:** Reliable committed events delivered to external systems after database commit;

15. Final implementation directive

Elveriq's mission is **not to clone a vendor's interface or chase a perfect feature count**. It is to build warehouse software that can safely direct real people and goods with the breadth to grow into more complex operations. The practical sequence is: transaction correctness -> bin inventory -> warehouse tasks -> scanning -> full picking/packing/dispatch -> 3PL customer workflows and billing -> labor/yard/optimization -> robotics and AI where customers justify it.

Treat the 300-line CSV register as the engineering backlog source, this paper as the researched architecture and requirements record, and pilot evidence as the authority for deciding which features earn priority. A warehouse should be able to answer at any moment: **what stock do we have, who owns it, where is it, what work is next, what went wrong, who confirmed it, what is leaving, and what service should be billed?**

Document revision: Version 1.0, research snapshot 12 November 2026. Feature presence and release packaging should be reconfirmed before external competitive claims or regulated customer commitments.